COMPASS_Synoptic_LE_Soil-Fluxes_2023_CH4

Roberta Peixoto

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1 See if we want to remove these fluxes or if we want to try and recalculate these ones? 22

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0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data

a. i.e. drops fluxes that have the drop column TRUE or FLAG this
b. adjusting times as needed

4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Unfortunatly - we had many problems with labels - we need to include the sites in the Plot_ID

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
[1] "CSV file exists and has been read into the code."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

```
1 entry had no timestamp matches!
```

0.8 fixing some Deadbands Revisit here, it is not working to fix the dead band

```
Warning in left_join(., time_overrides, by = "Plot_ID"): Detected an unexpected many-
to-many relationship between `x` and `y`.
i Row 30585 of `x` matches multiple rows in `y`.
i Row 16 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

##Extracting initial CH4 concentration

0.9 Change the units from ppb to nmol

```
#change the units
alldat_trimmed$CH4_nmol <- ffi_ppb_to_nmol(alldat_trimmed$ch4,
                                volume = (alldat_trimmed$TotalVolume_fixed_cm3 / 1000000), #
                                temp = alldat_trimmed$chamber_t)    # degrees C
```

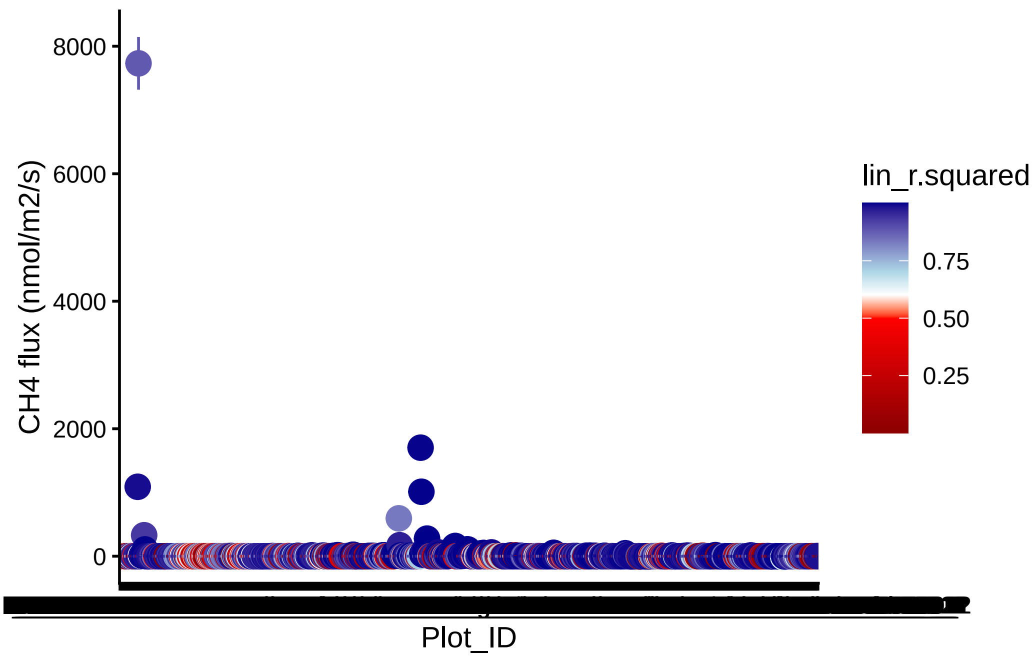
```
Assuming atm = 101325 Pa
```

```
Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

```
#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1

alldat_trimmed$CH4_nmol_m2 <- alldat_trimmed$CH4_nmol / 0.0318 #adjust for the area
```

0.10 Calculate fluxes

0.11 Visualize fluxes and sort by r2

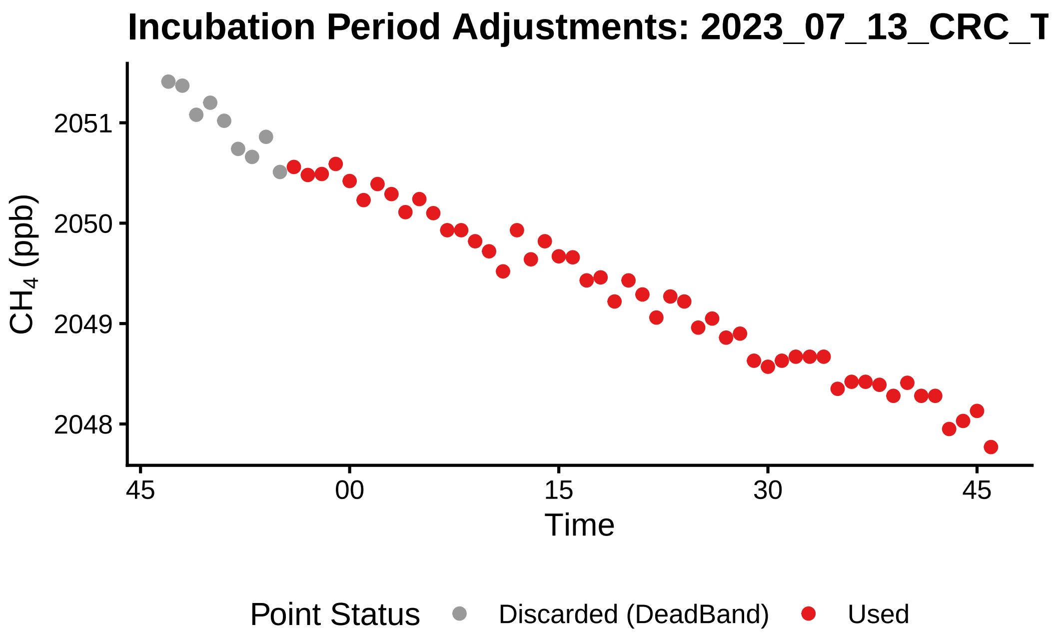


0.12 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

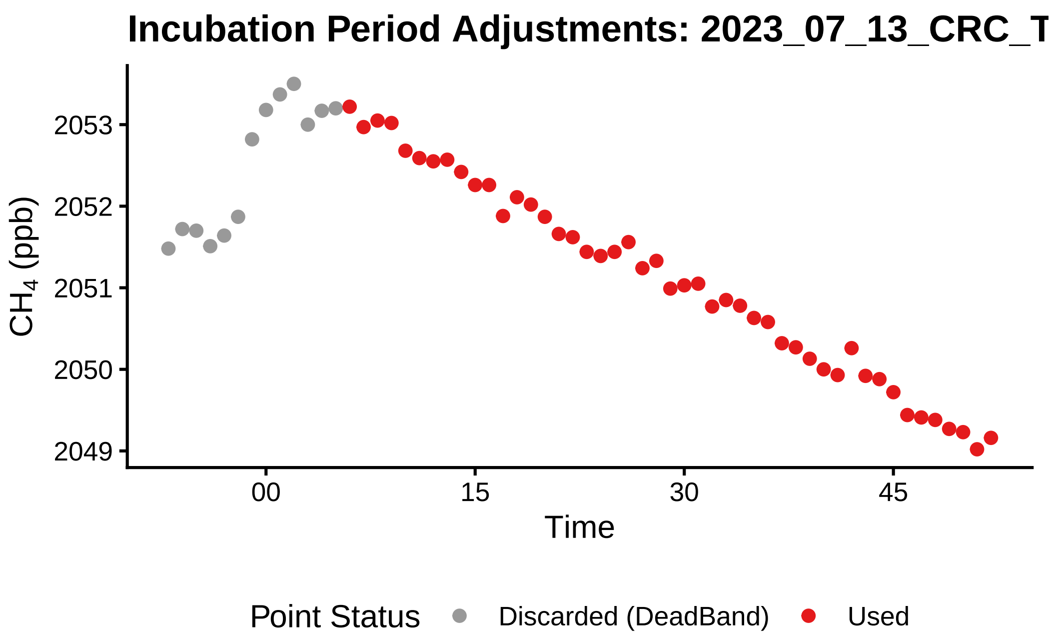
```
# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  group_by(Plot_ID) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP),
    .groups = "drop"
  )

# use full alldat + trim limits to classify each point
r_all <- alldat %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  left_join(trim_limits, by = "Plot_ID") %>%
```

```r
  group_by(Plot_ID) %>%
  mutate(
    time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
    point_status = case_when(
      TIMESTAMP < t_start_trim ~ "Discarded (DeadBand)",
      TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
      TRUE                     ~ "Used"
    )
  ) %>%
  ungroup()

# ---- PLOTTING: one plot per Plot_ID (PDF friendly) ----

plot_ids <- unique(r_all$Plot_ID)

for (P in plot_ids) {

  p <- ggplot(
    r_all %>% filter(Plot_ID == P),
    aes(x = TIMESTAMP, y = ch4, color = point_status)
  ) +
    geom_point(size = 1.8) +
    scale_color_manual(
      values = c(
        "Used"                  = "#E41A1C",
        "Discarded (DeadBand)"  = "grey60",
        "Discarded (stop trim)" = "grey20"
      )
    ) +
    theme_classic(base_size = 12) +
    ylab(expression(paste(CH[4], " (ppb)"))) +
    xlab("Time") +
    labs(
      title = paste("Incubation Period Adjustments:", P),
      color = "Point Status"
    ) +
    theme(
      legend.position = "bottom",
      plot.title = element_text(face = "bold")
    )

  print(p)   # IMPORTANT: each print() = new page in PDF
```

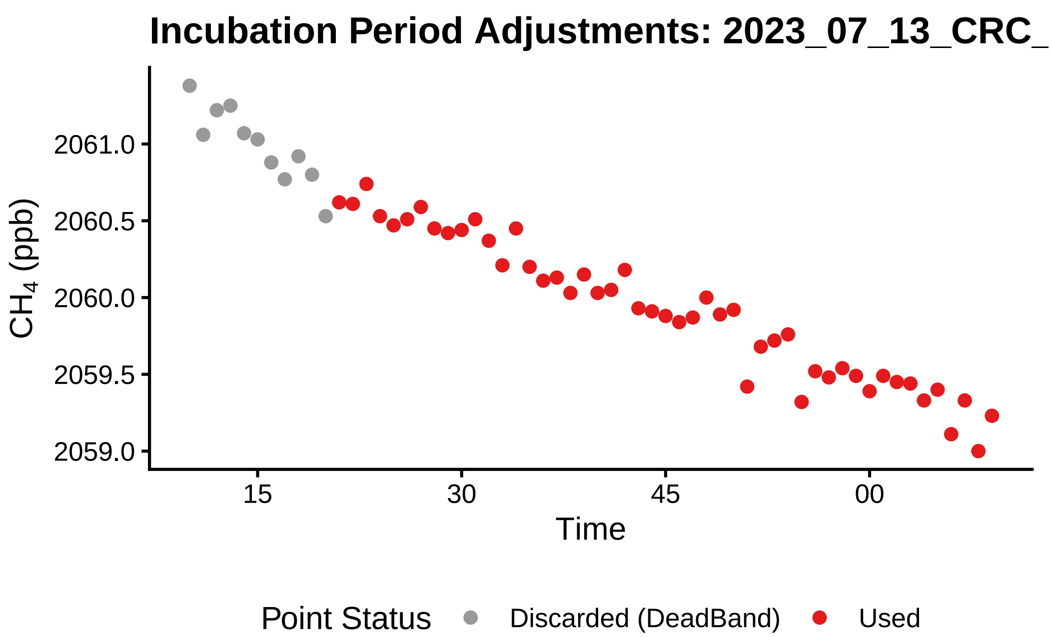
```
}
```

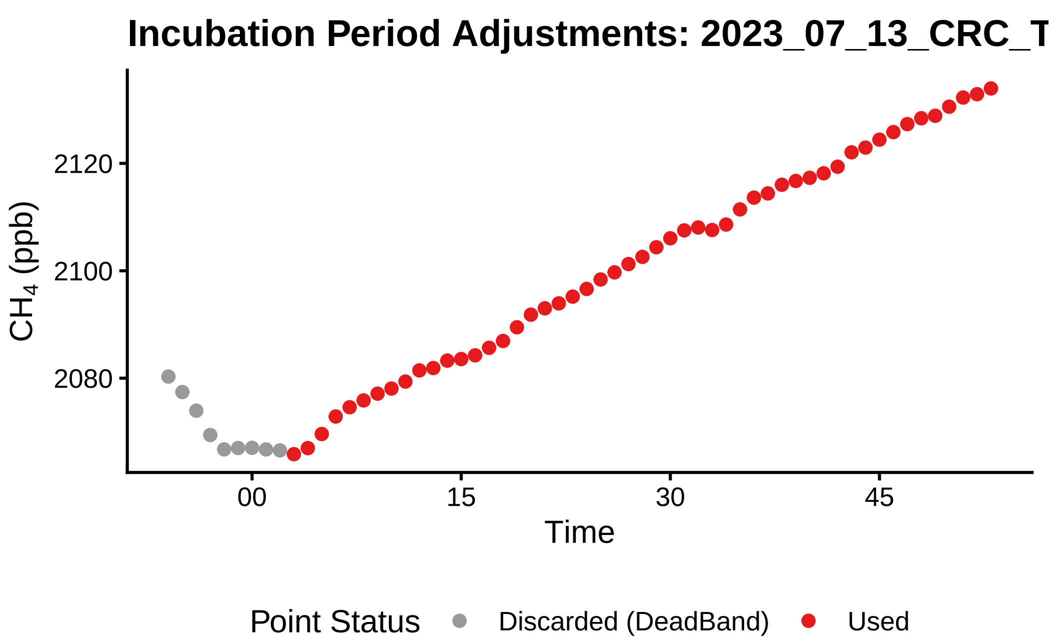


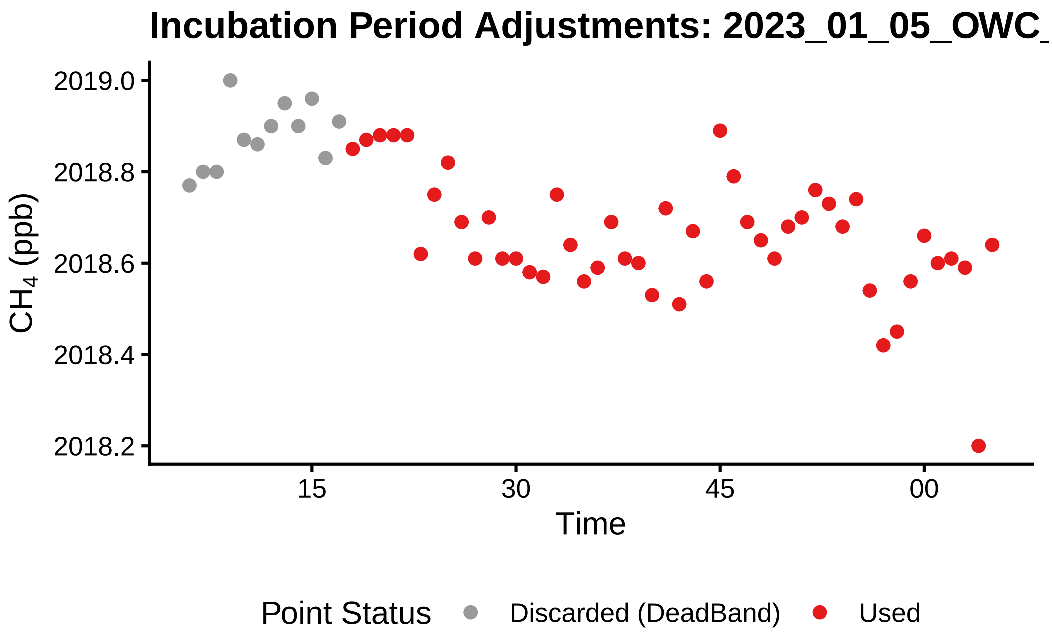
Incubation Period Adjustments: 2023_07_13_CRC_T
2051
2050
2049
2048
CH4 (ppb)
45
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

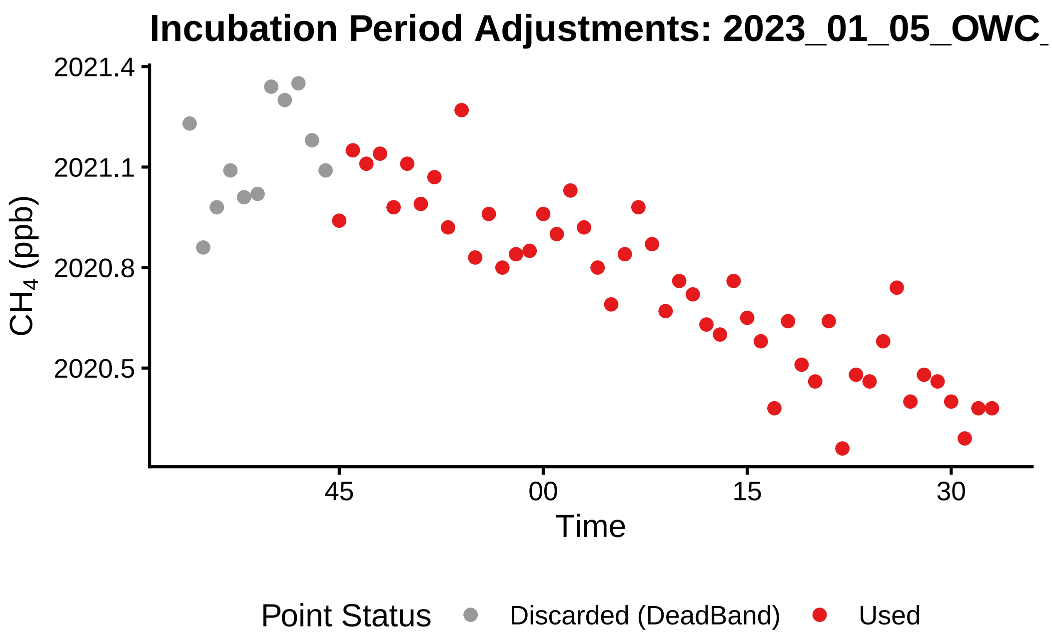


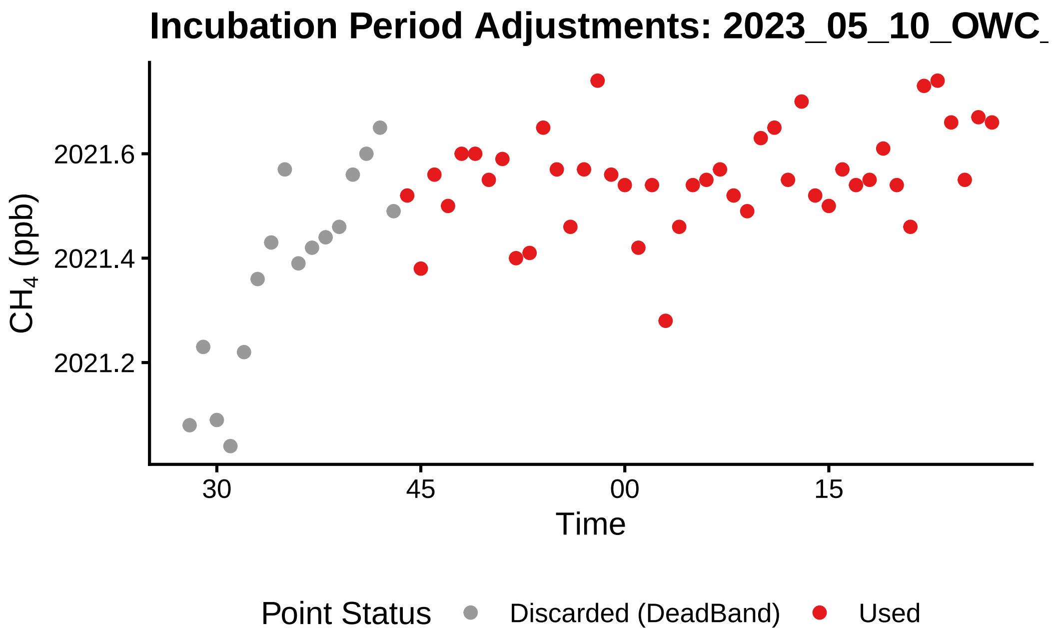
Incubation Period Adjustments: 2023_07_13_CRC_T
2053
2052
2051
2050
2049
CH4 (ppb)
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

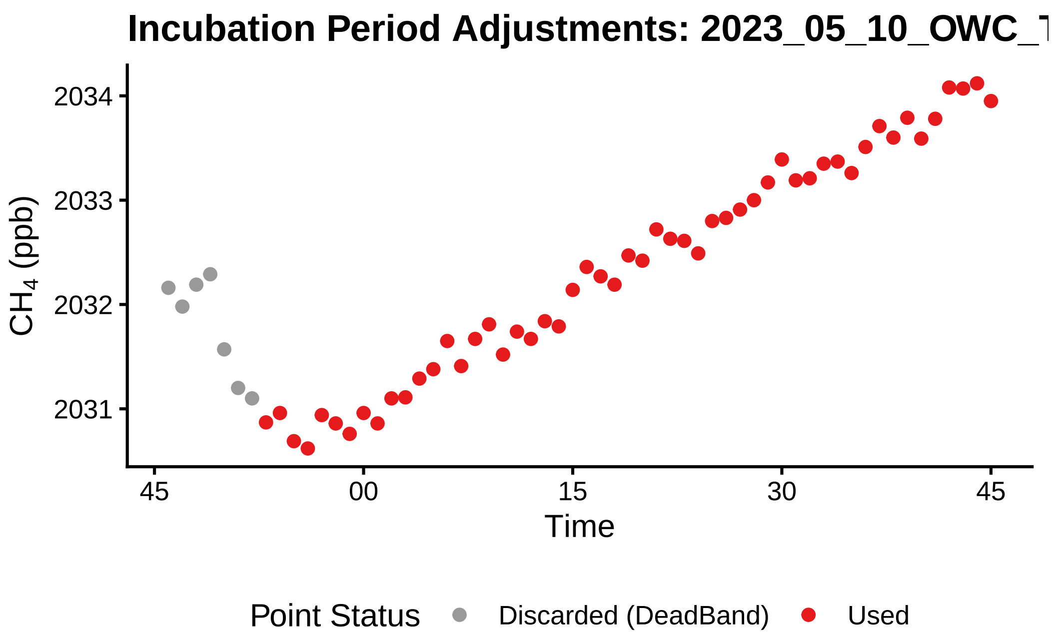

Incubation Period Adjustments: 2023_07_13_CRC_
2061.0
2060.5
2060.0
2059.5
2059.0
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

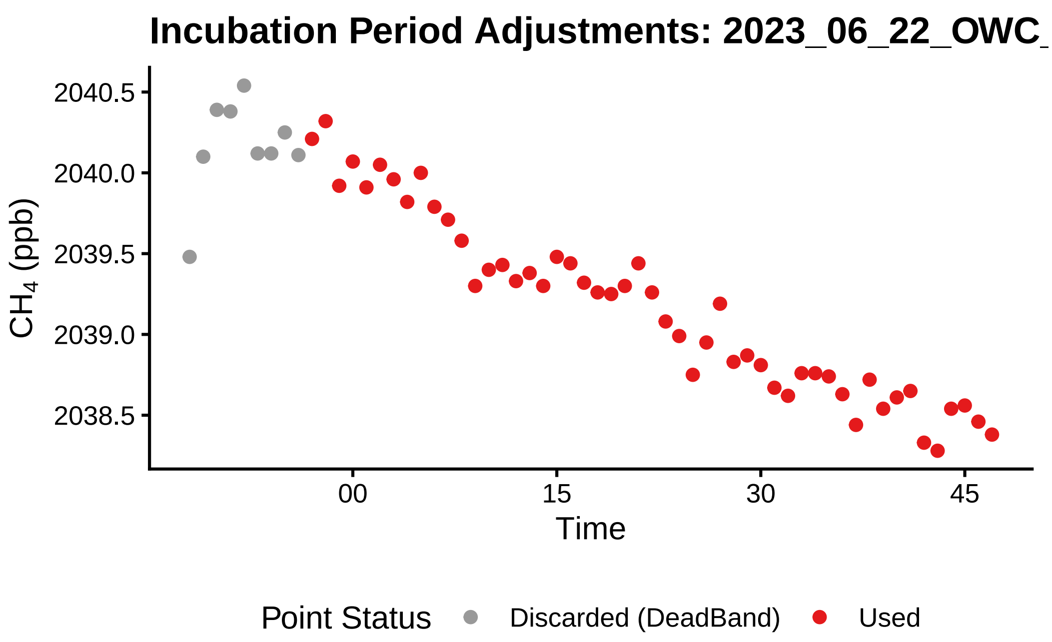

Incubation Period Adjustments: 2023_07_13_CRC_T
2120
2100
2080
CH4 (ppb)
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

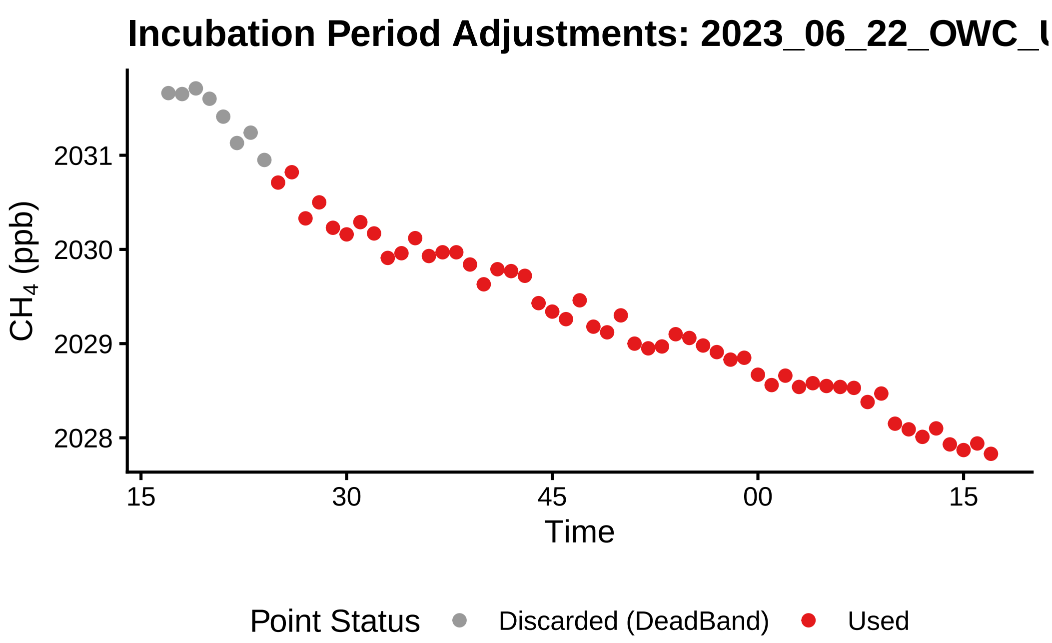

Incubation Period Adjustments: 2023_01_05_OWC_
2019.0
2018.8
2018.6
2018.4
2018.2
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

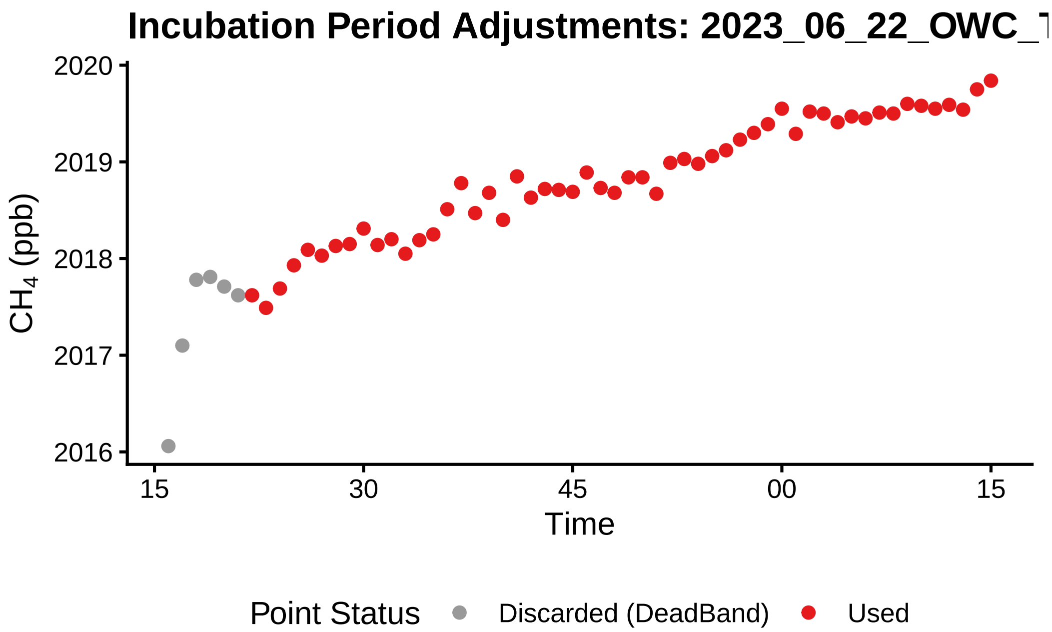

Incubation Period Adjustments: 2023_01_05_OWC_
2021.4
2021.1
2020.8
2020.5
CH4 (ppb)
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used

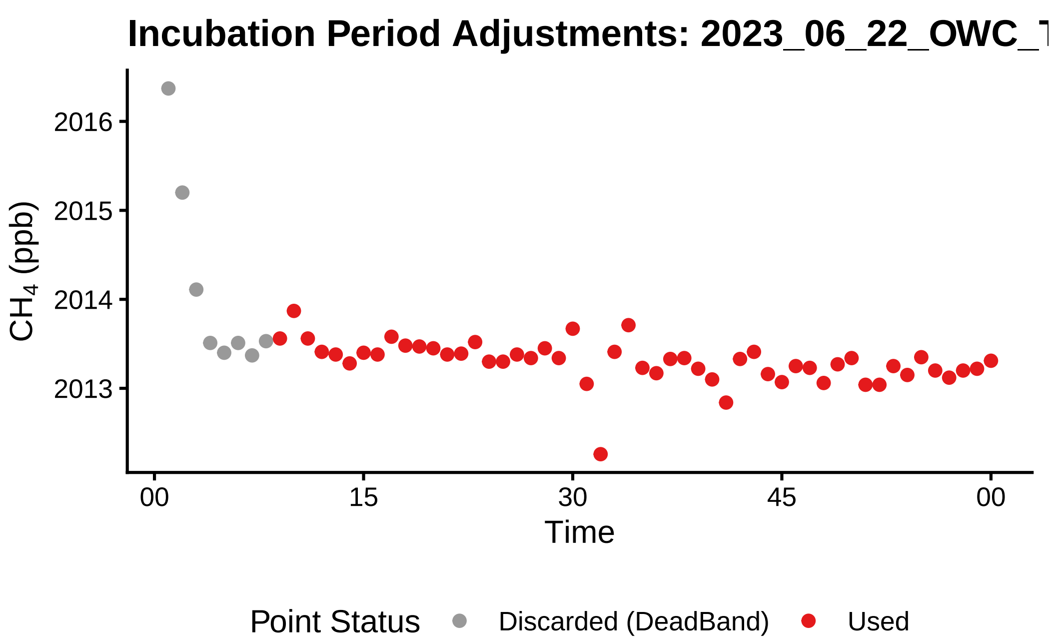

Incubation Period Adjustments: 2023_05_10_OWC_
2021.6
2021.4
2021.2
CH_4 (ppb)
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

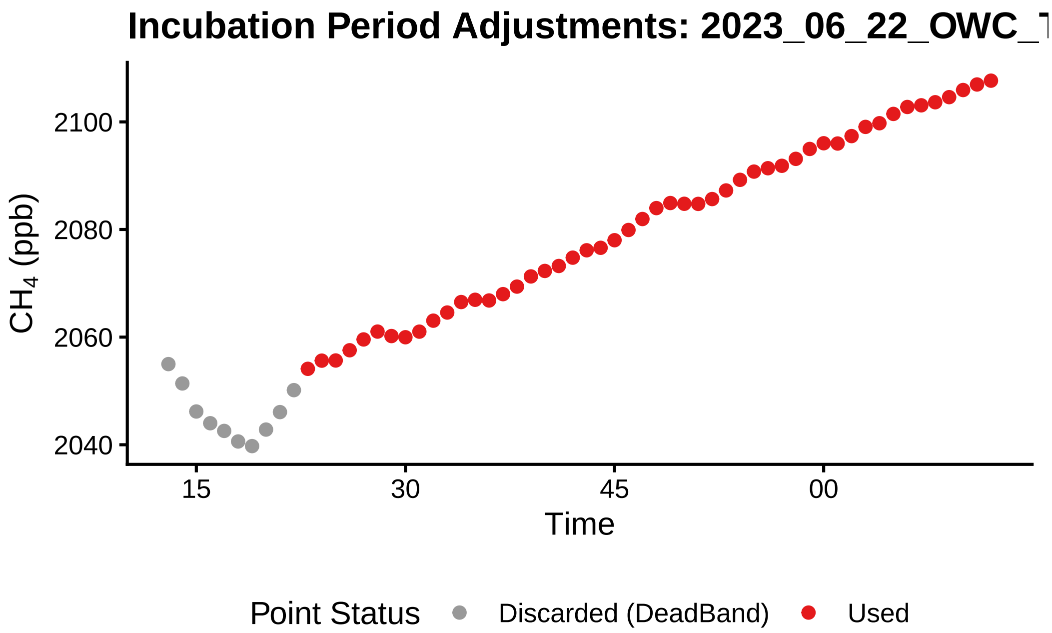

Incubation Period Adjustments: 2023_05_10_OWC_
2034
2033
2032
2031
CH_4 (ppb)
45
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2023_06_22_OWC_
2040.5
2040.0
2039.5
2039.0
2038.5
CH4 (ppb)
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

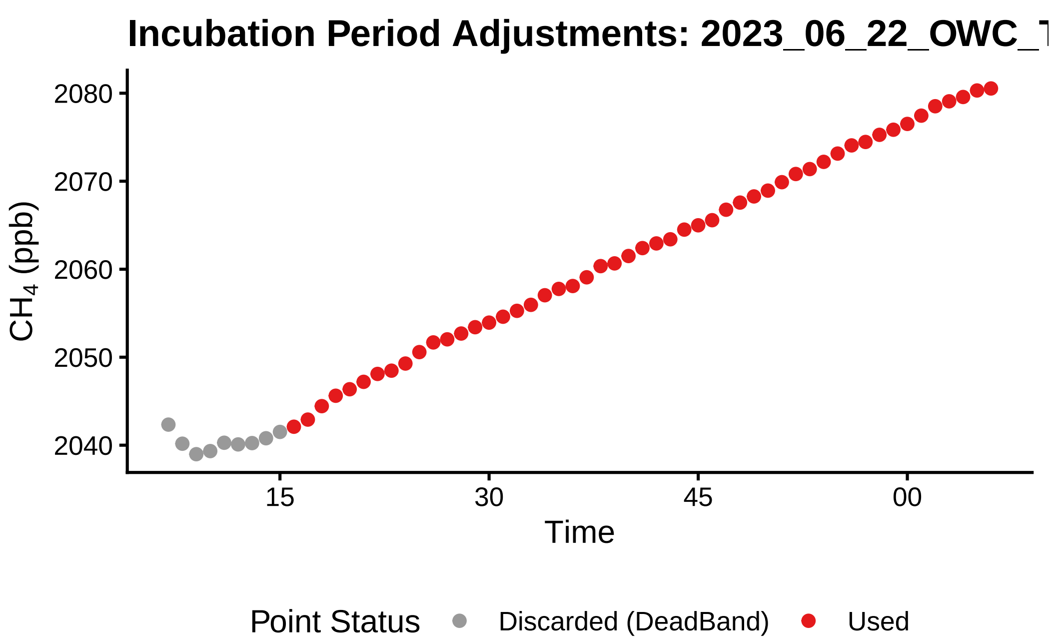

Incubation Period Adjustments: 2023_06_22_OWC_U
2031
2030
2029
2028
CH4 (ppb)
15
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
2020
2019
2018
2017
2016
15
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

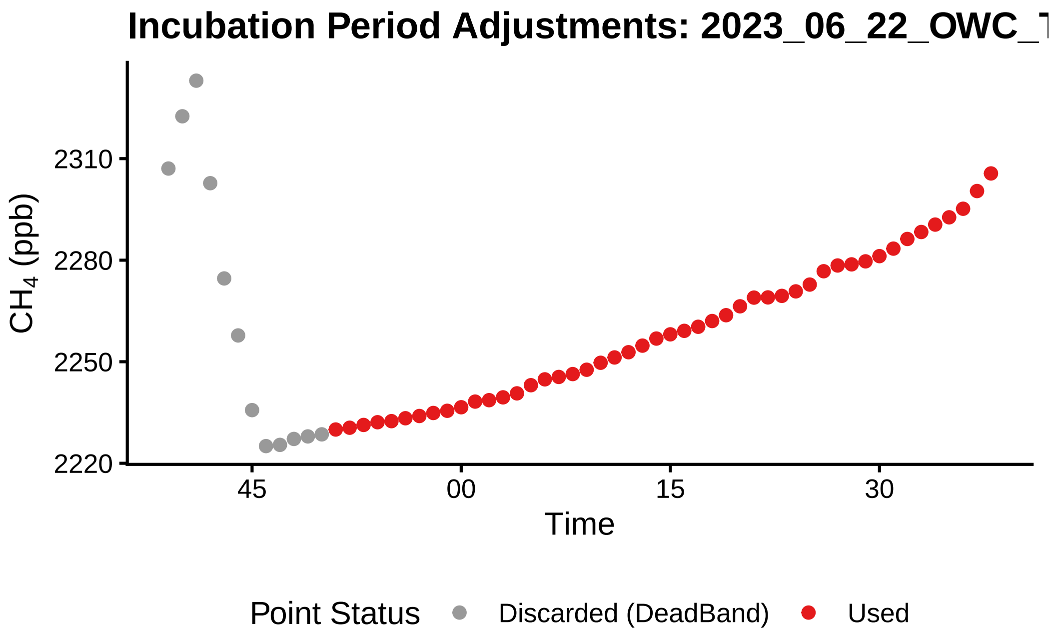

Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
2016
2015
2014
2013
00
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

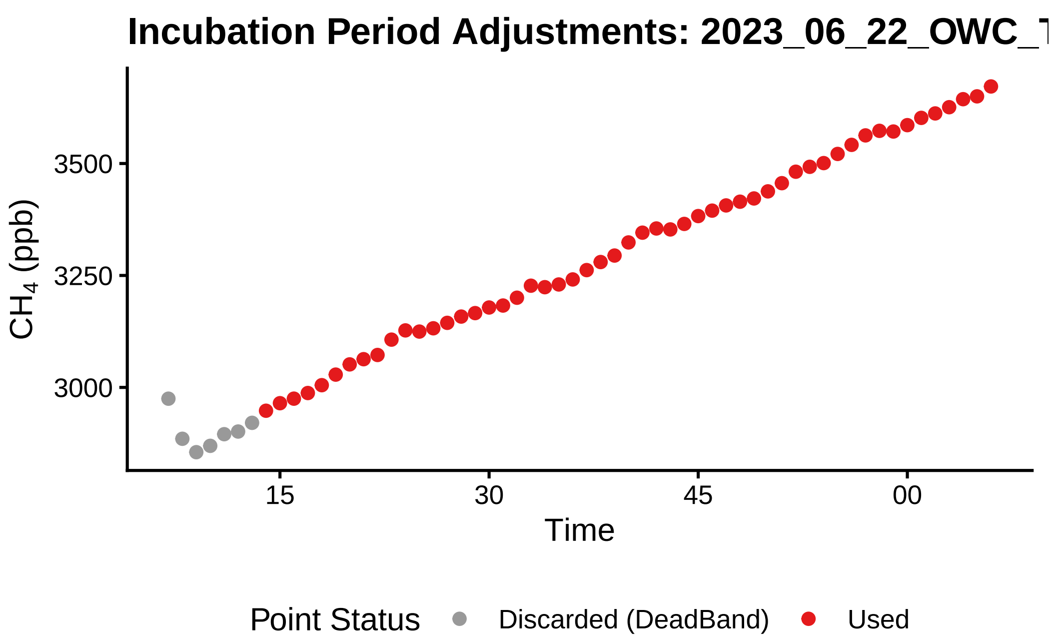


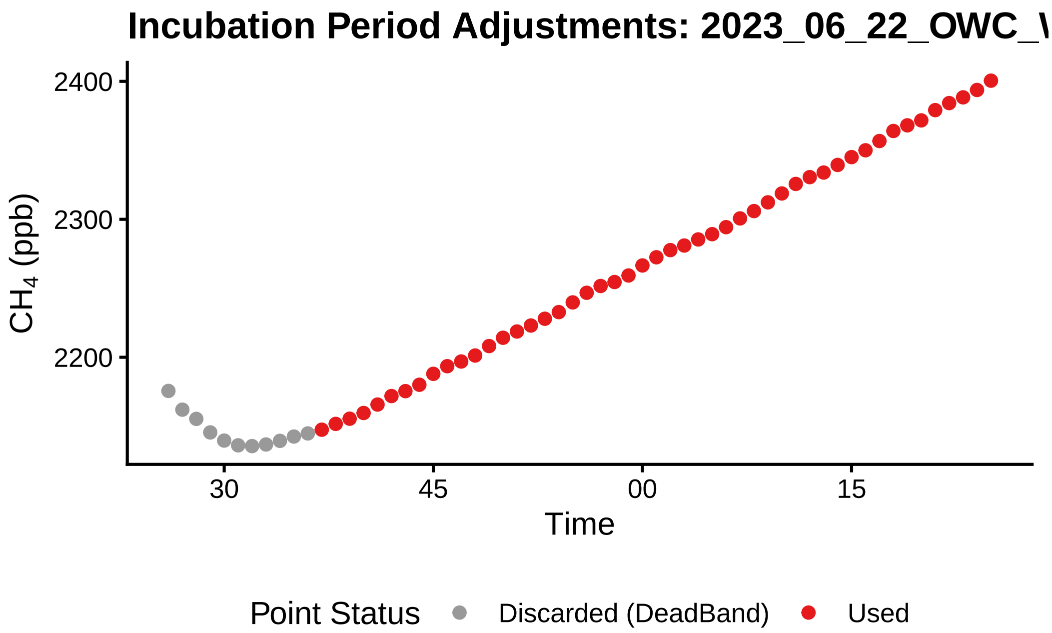
Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
2100
2080
2060
2040
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

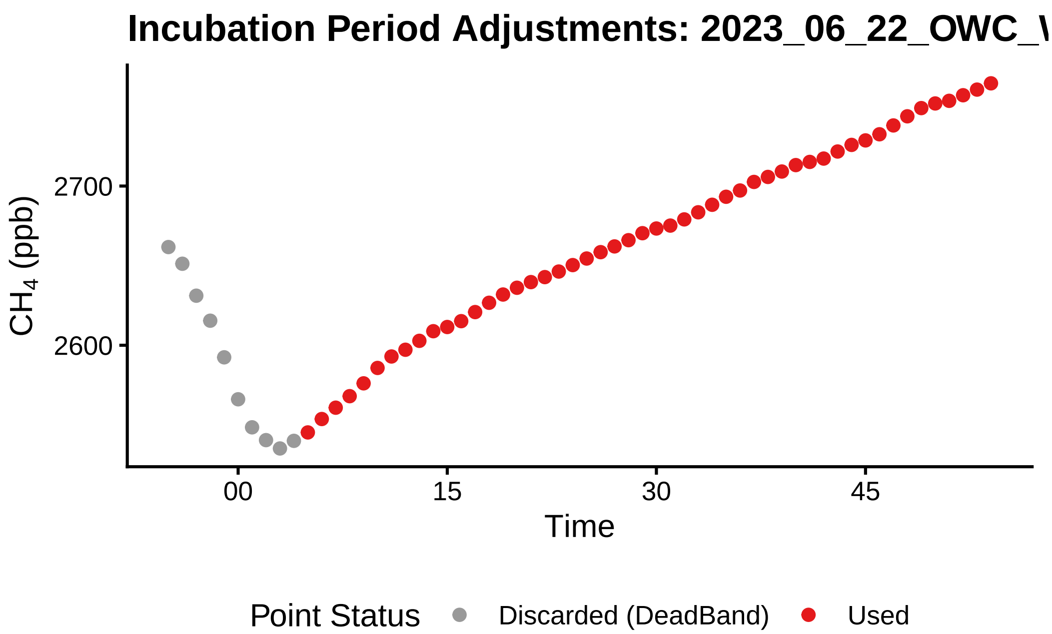


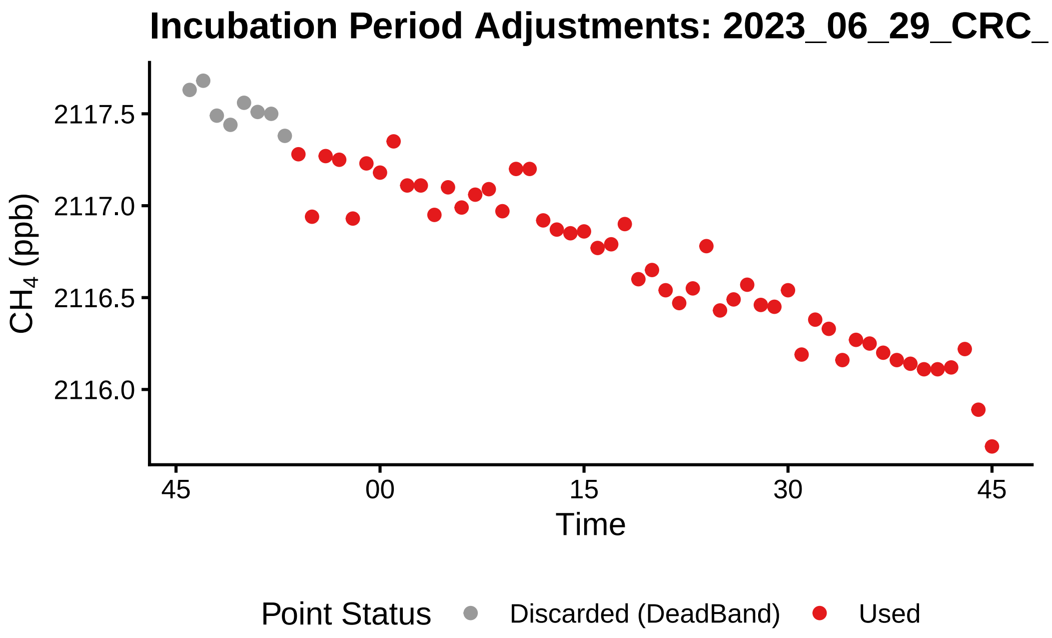
Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
2080
2070
2060
2050
2040
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

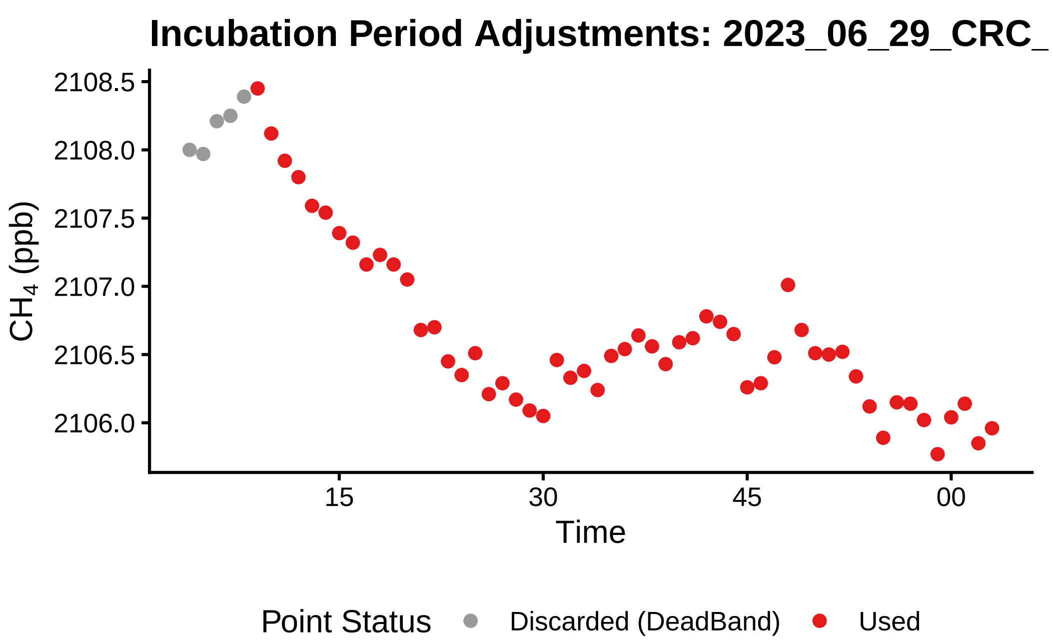

Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
2310
2280
2250
2220
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used

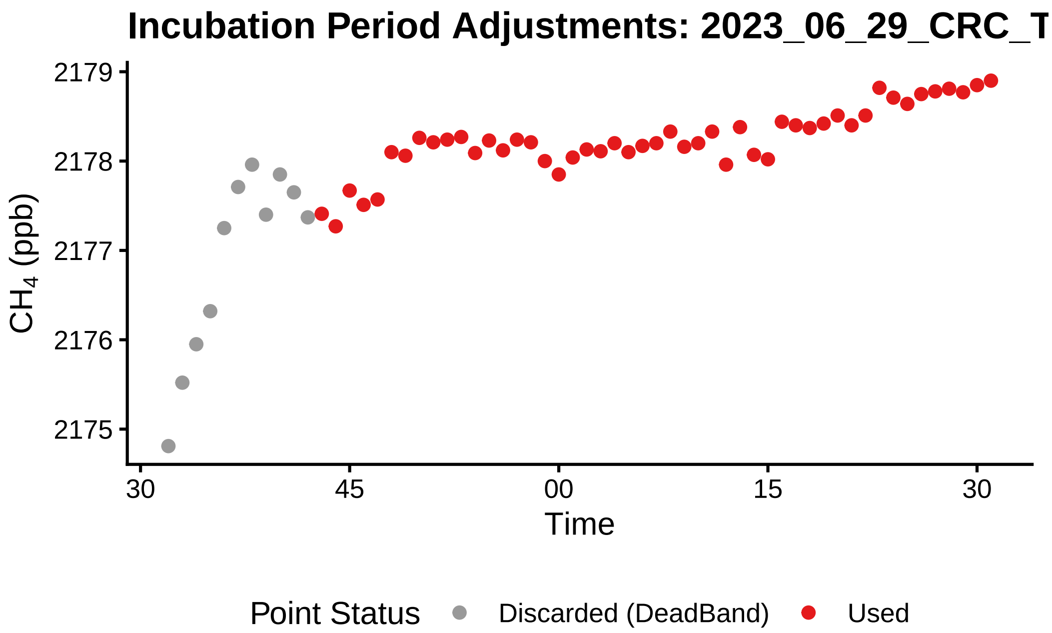

Incubation Period Adjustments: 2023_06_22_OWC_T
CH4 (ppb)
3500
3250
3000
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2023_06_22_OWC_\
CH4 (ppb)
2400
2300
2200
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2023_06_22_OWC_\
CH4 (ppb)
2700
2600
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

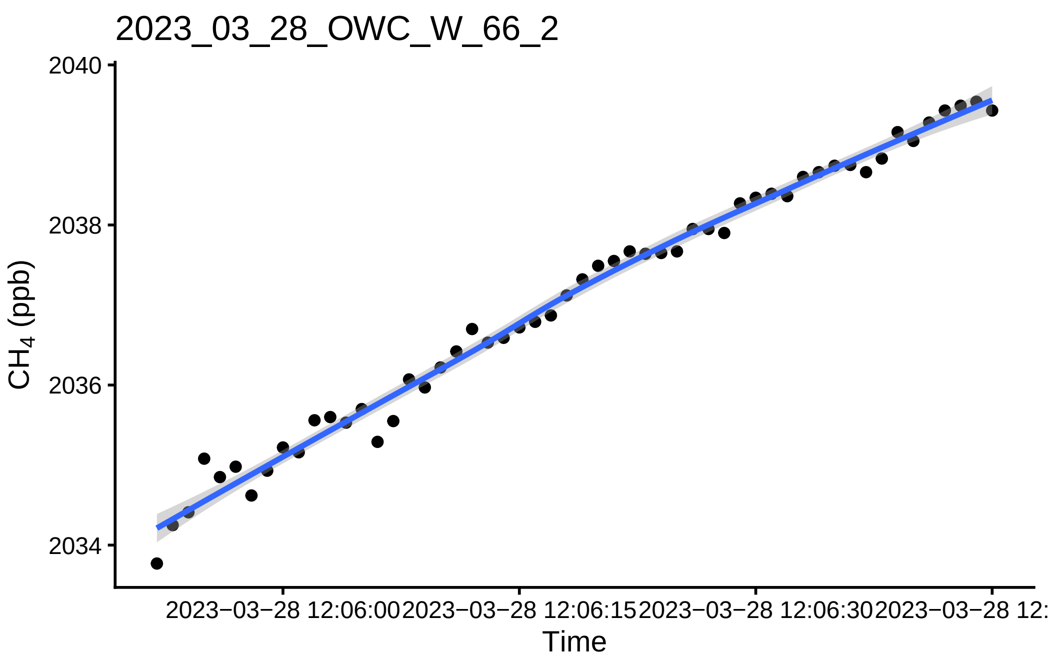

Incubation Period Adjustments: 2023_06_29_CRC_
CH$_4$ (ppb)
2117.5
2117.0
2116.5
2116.0
45
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2023_06_29_CRC_
CH$_4$ (ppb)
2108.5
2108.0
2107.5
2107.0
2106.5
2106.0
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

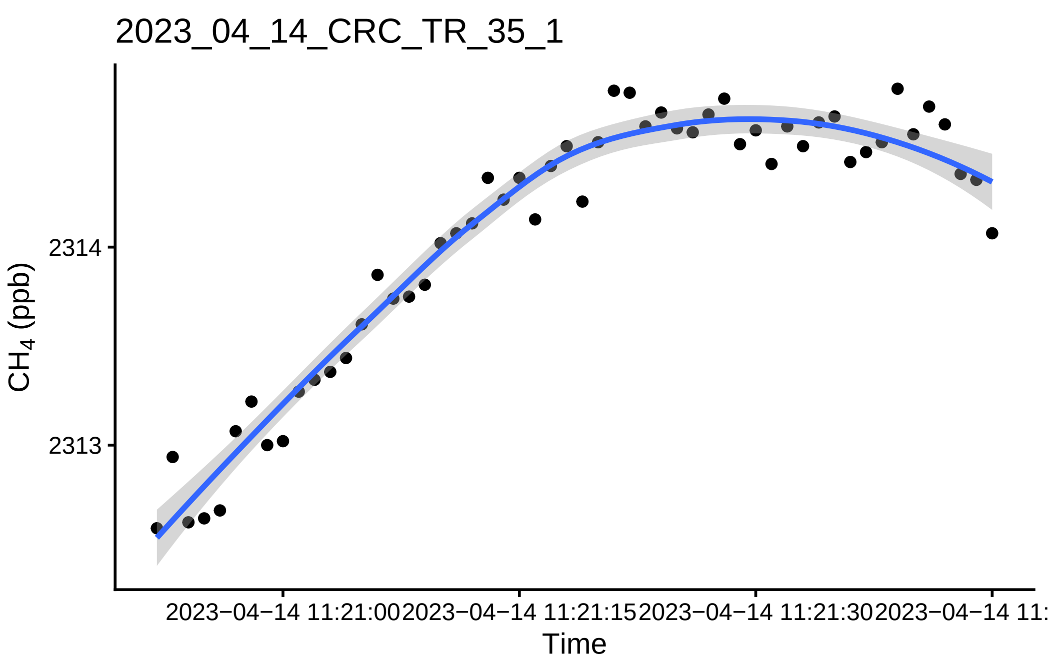


0.13 Spot check a few fluxes with R2 > 0.9

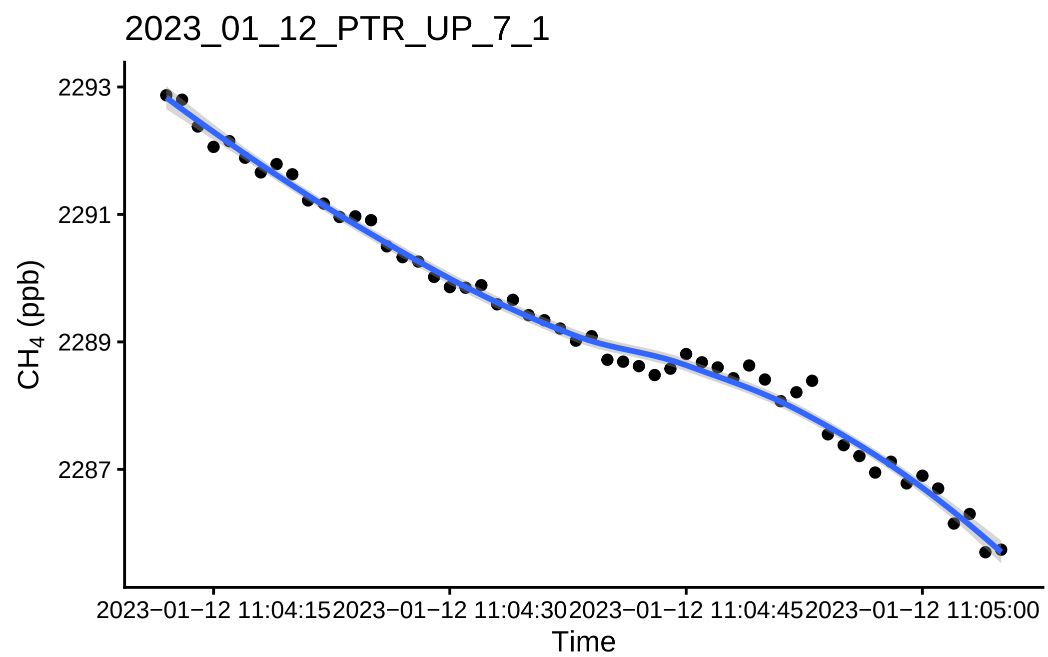
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



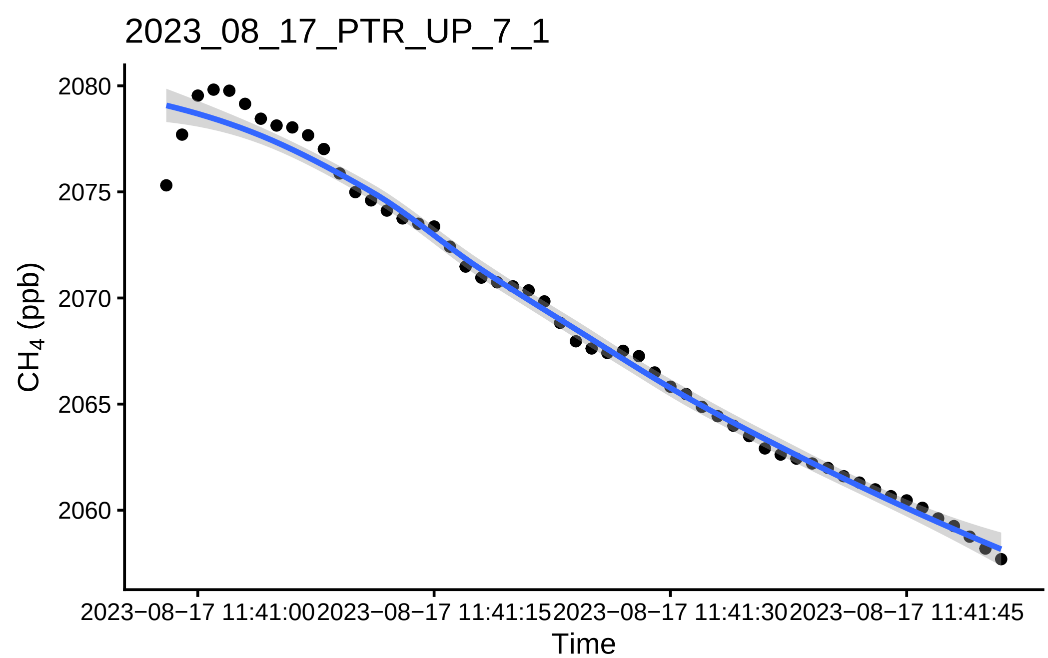

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



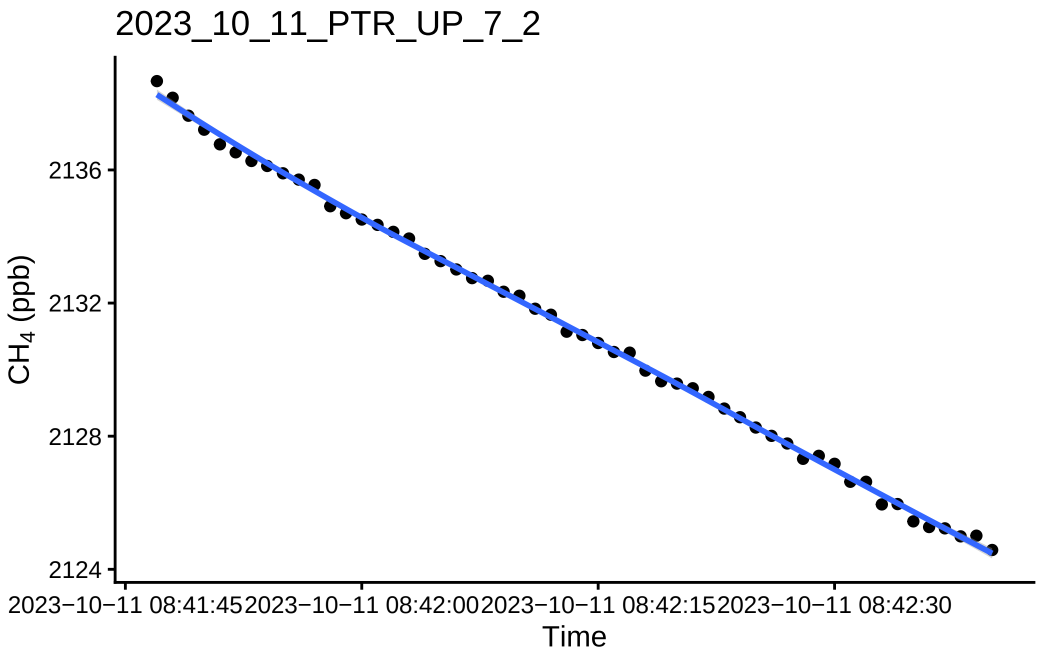

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

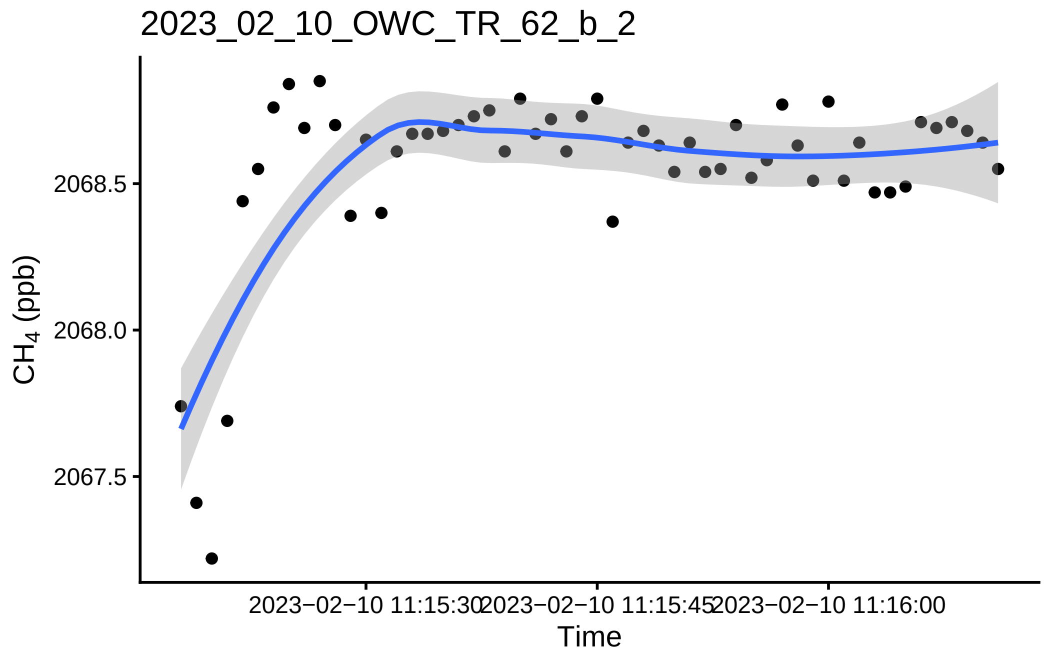



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

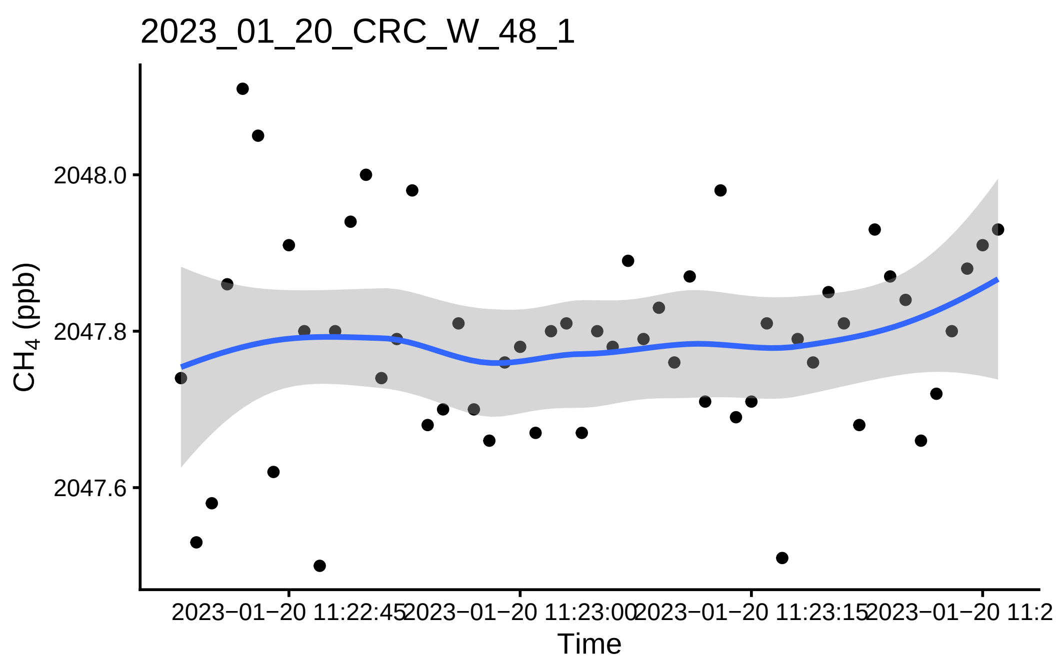


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

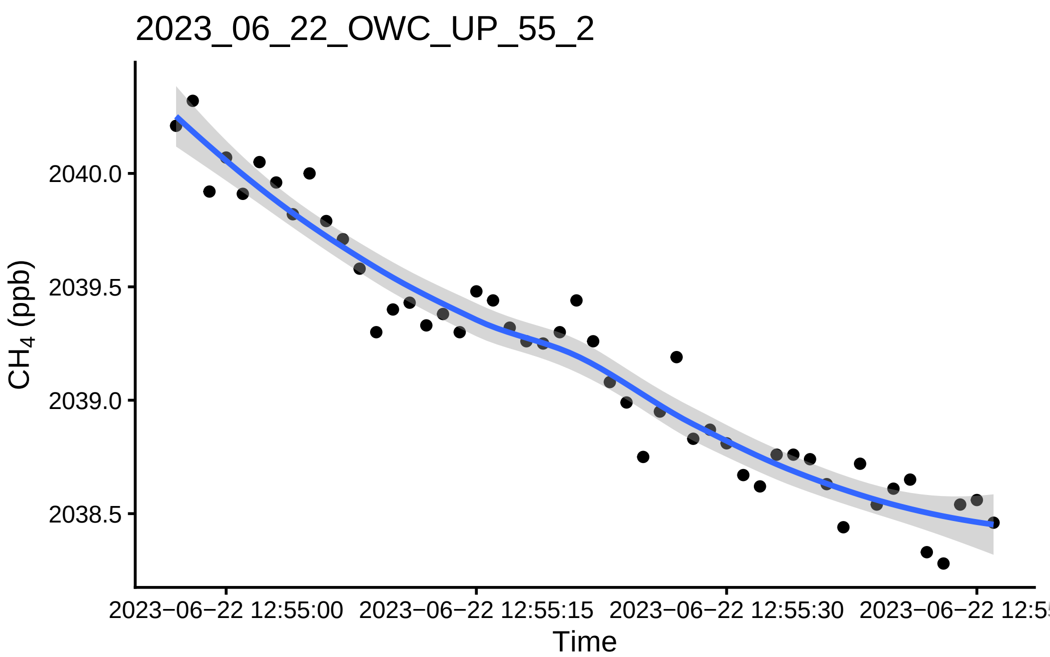
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



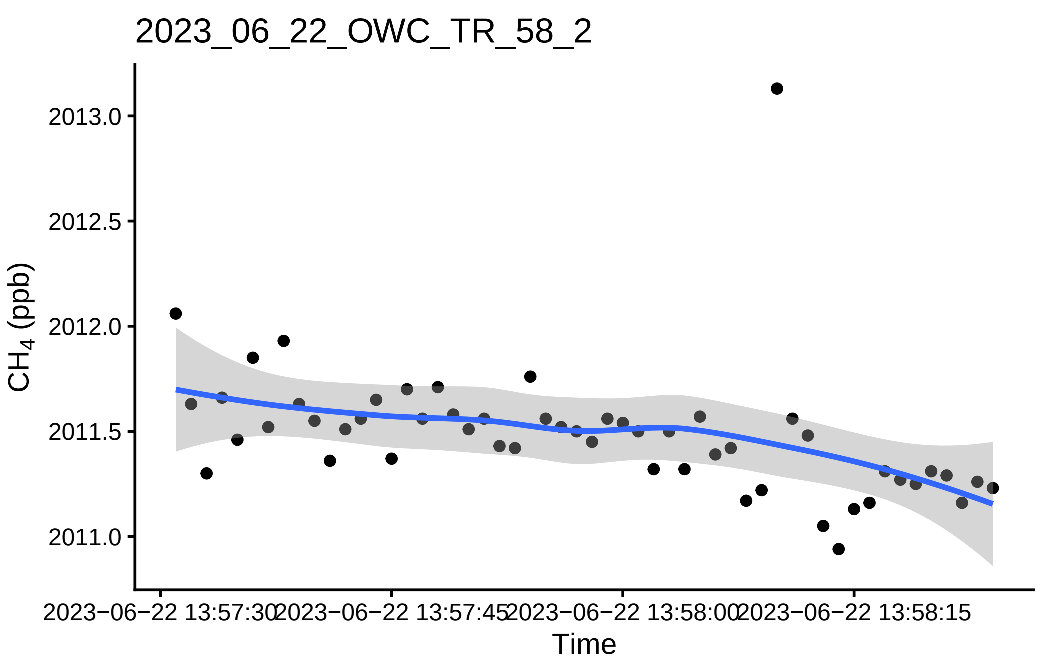
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



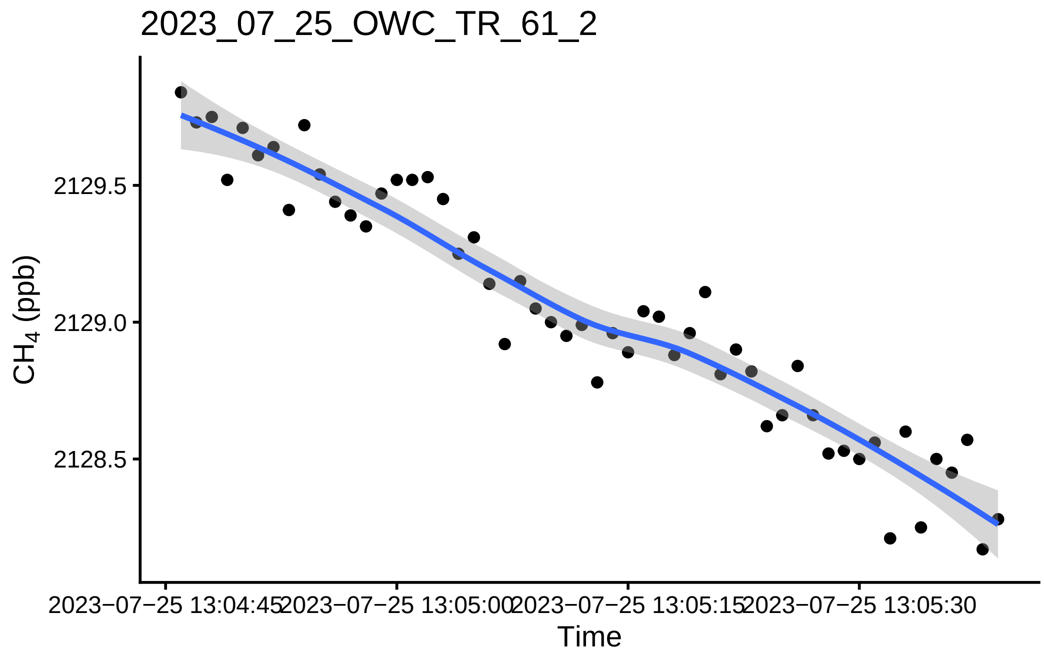
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



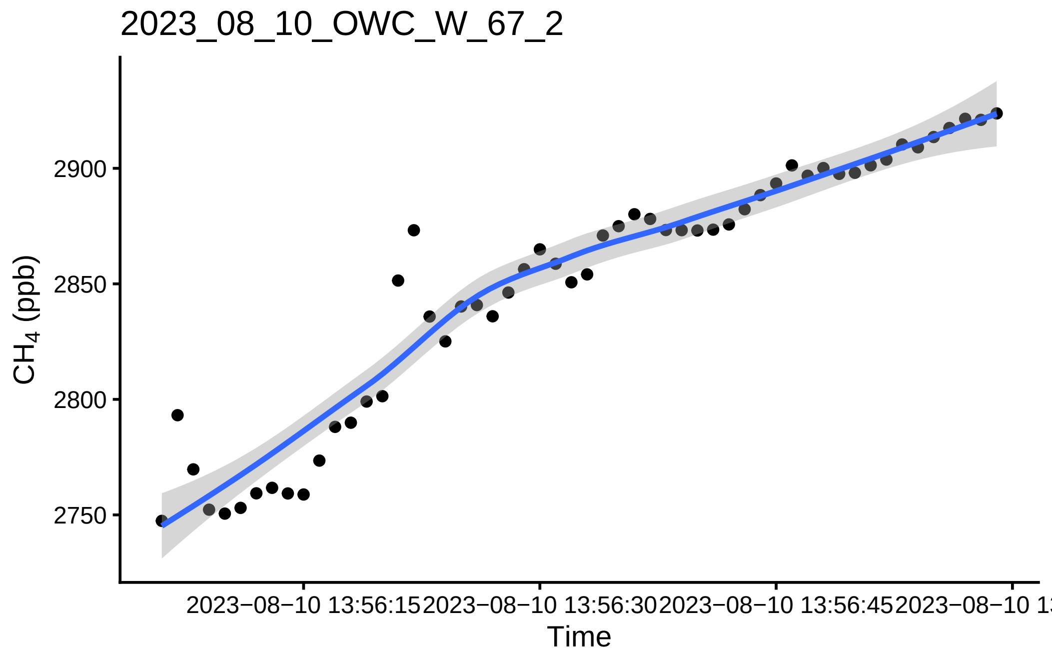
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



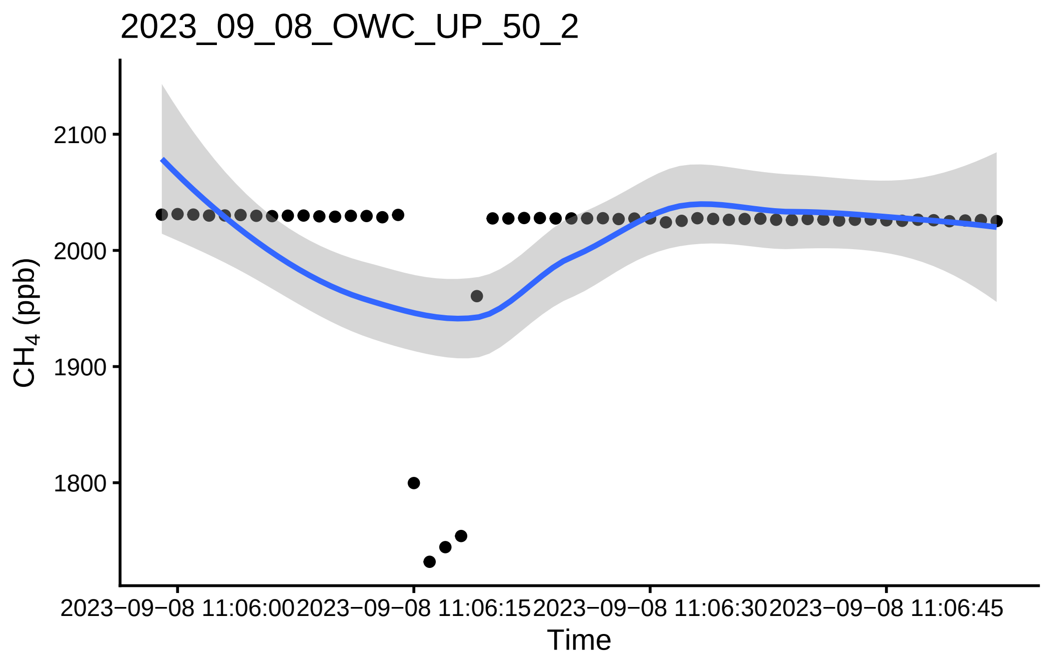
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

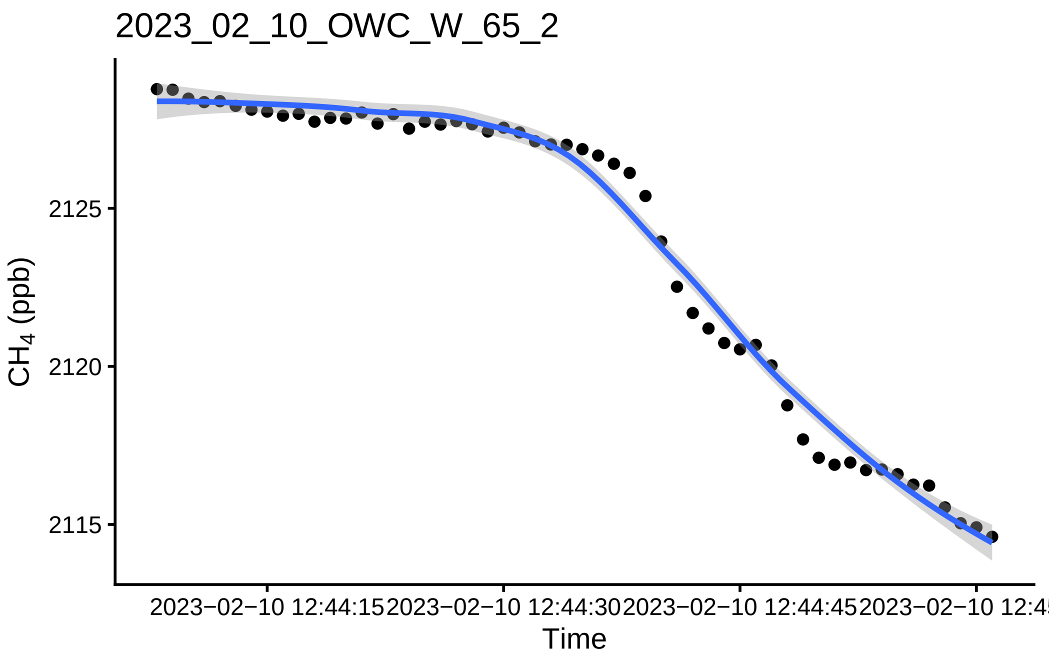



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

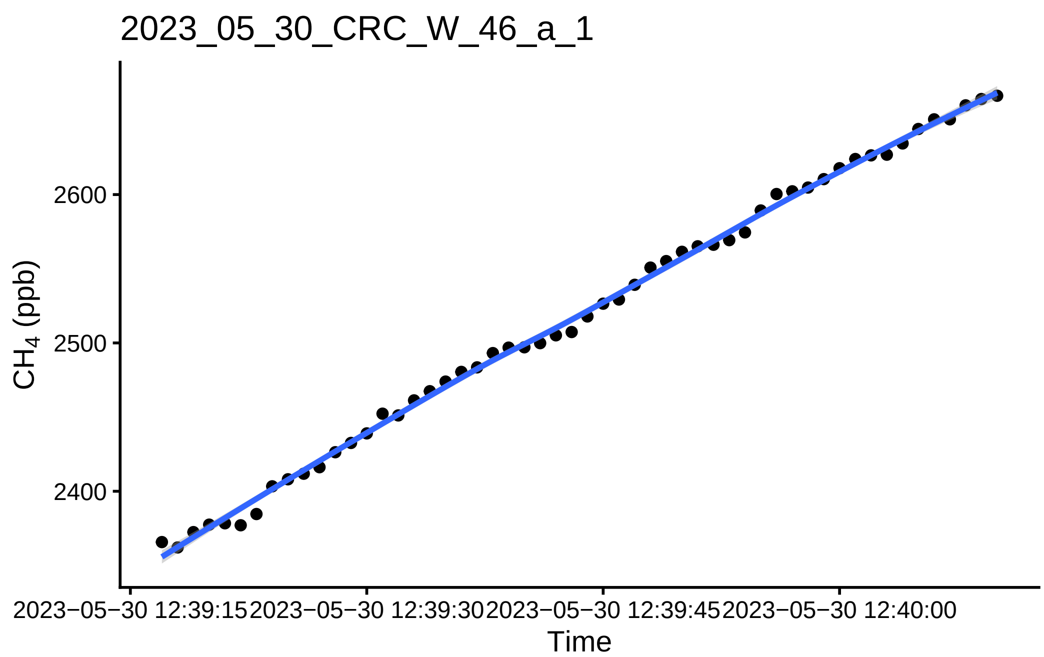


1.1 Look at the really high or really low ones to see if there is a timing issue or something

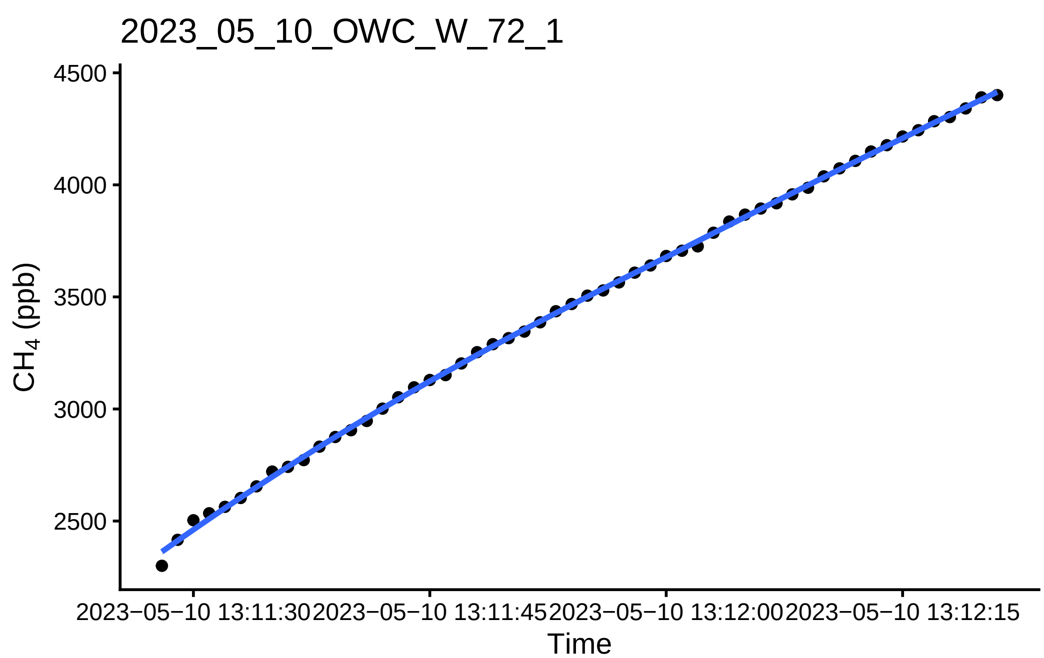
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



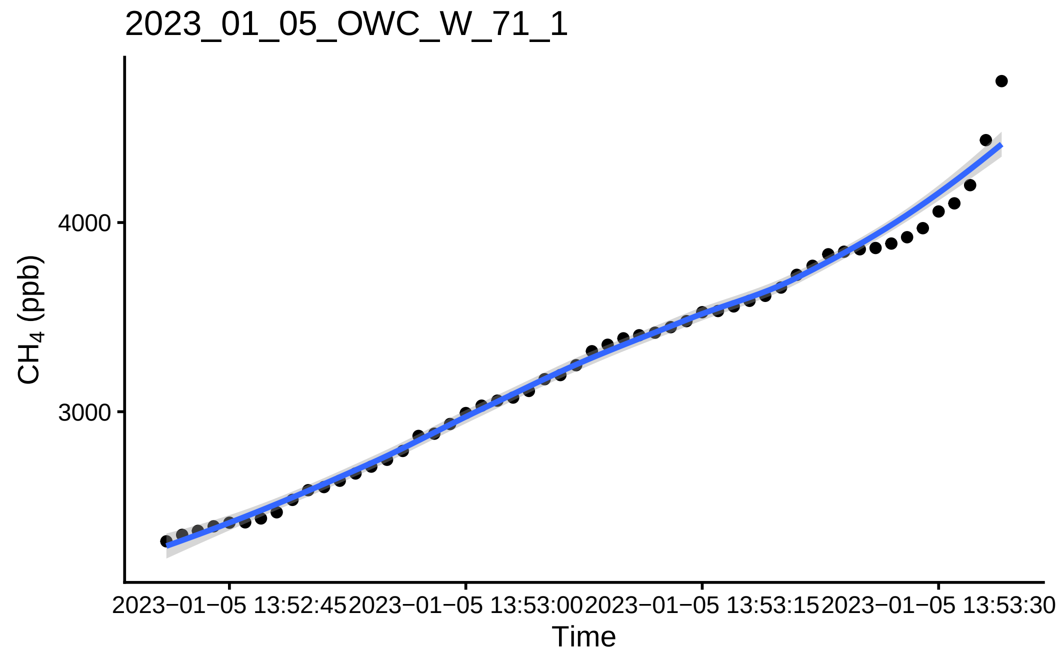

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



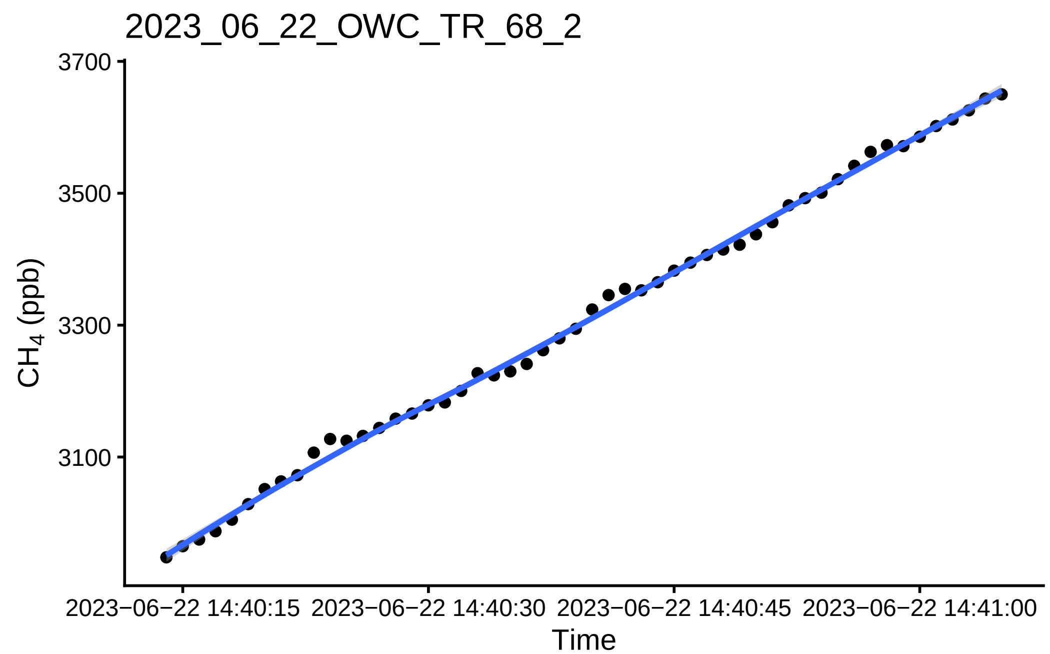

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

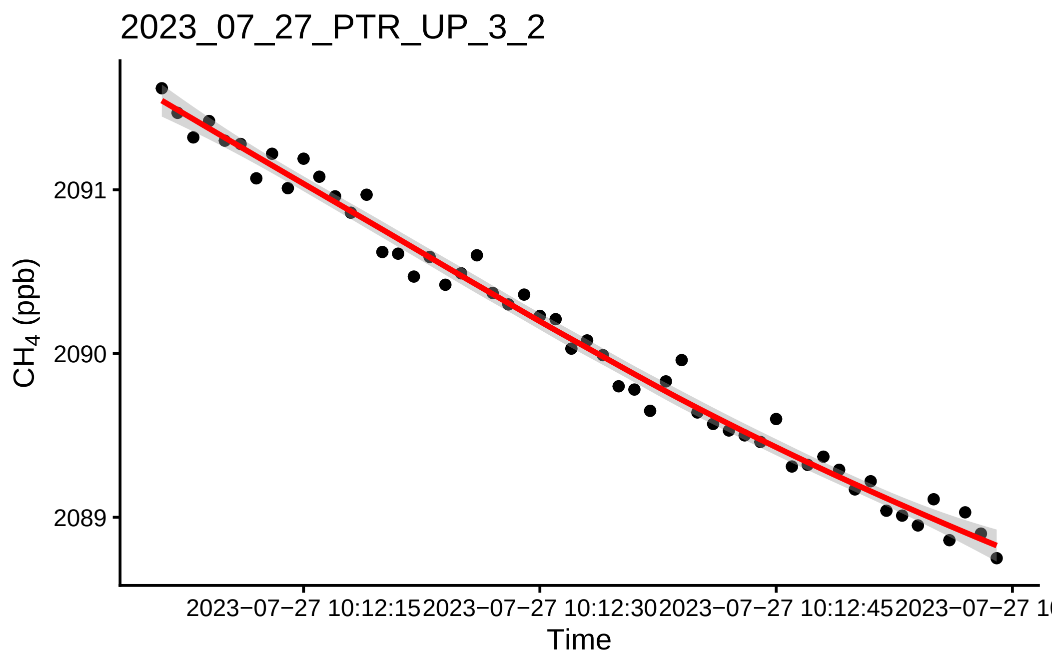



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

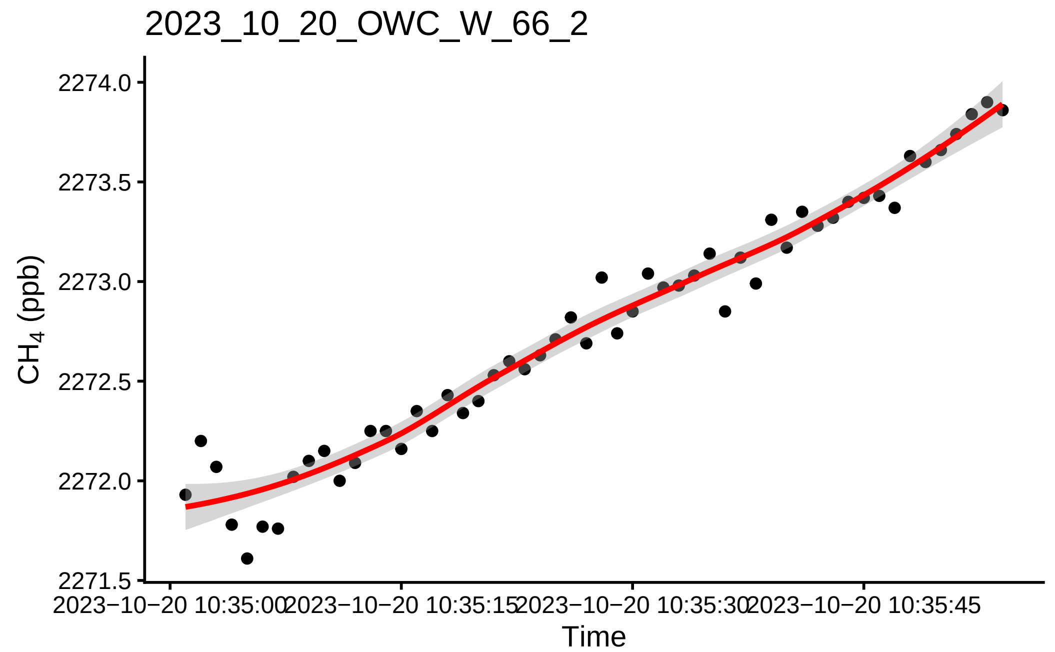


1.2 Check all fluxes

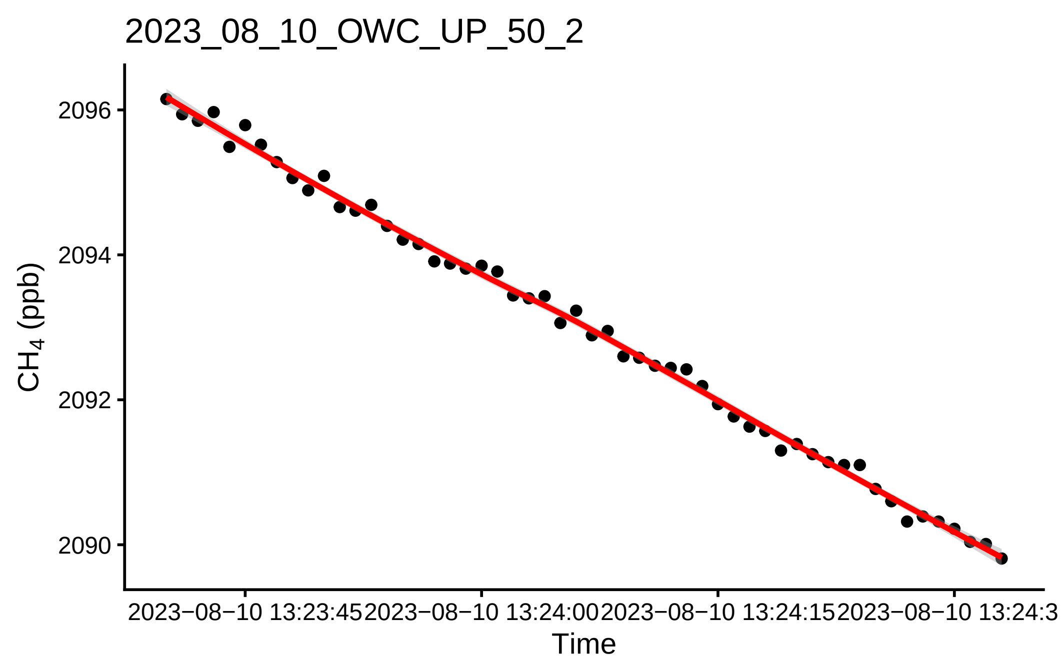
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



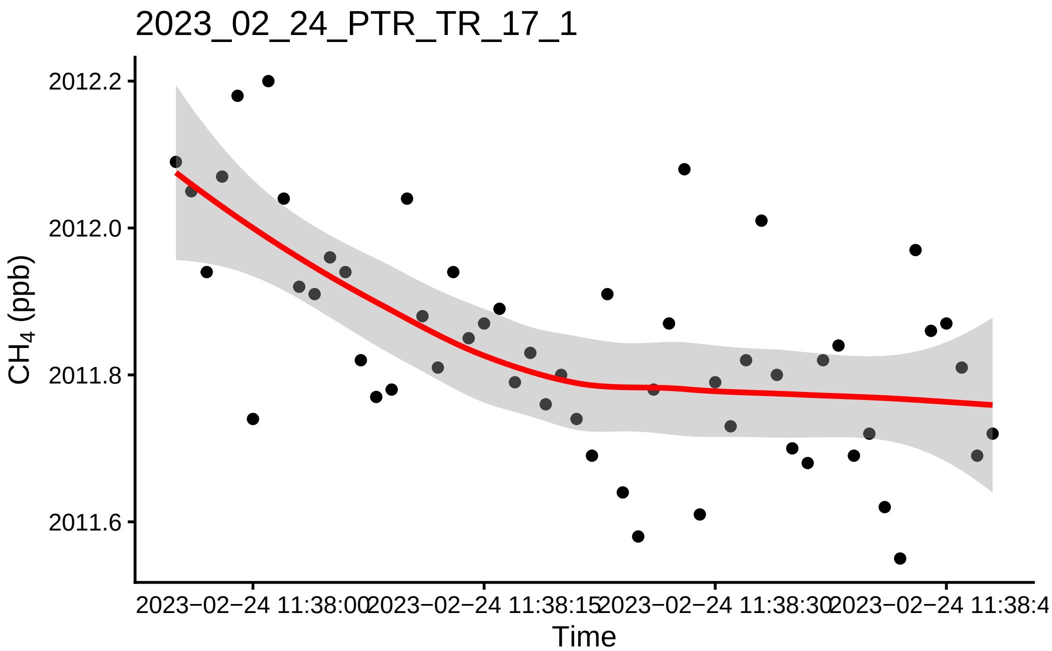

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



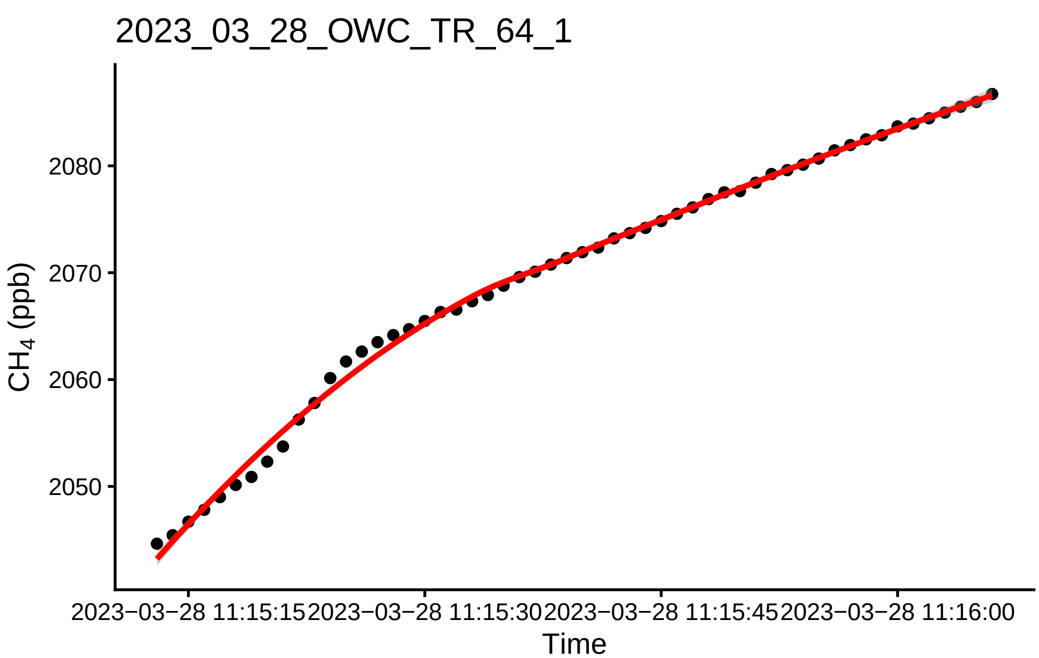

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



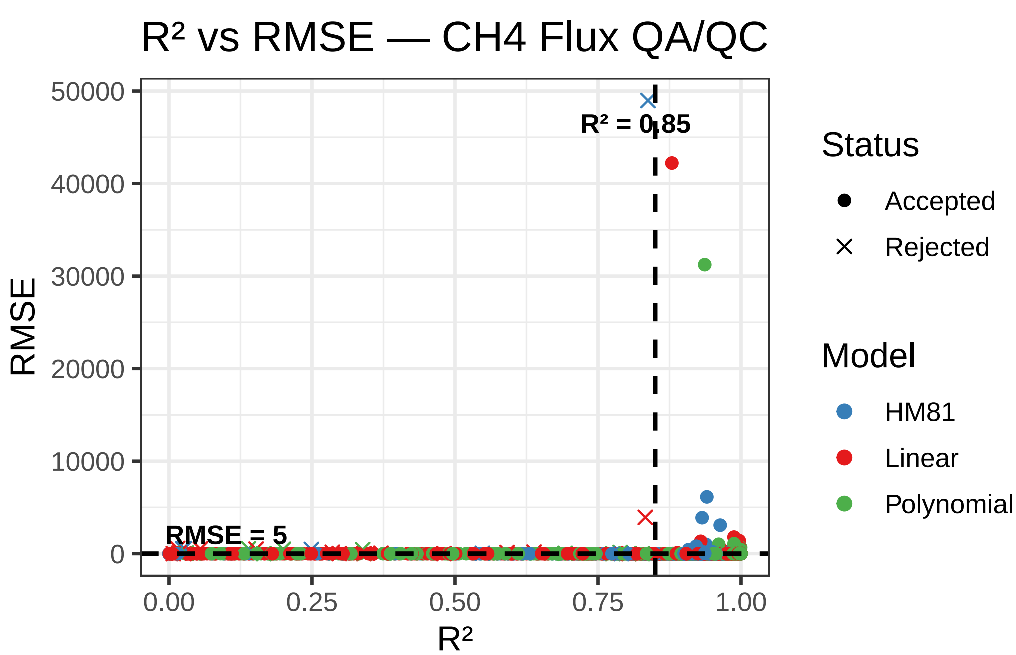

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



1.3 QAQC - Flags

Table 1: QA/QC CH4 Flux Summary

Metric	Value
Total observations	867
Rejections	
Rejected — poor model fit (n)	9
Rejected — model (%)	1
Rejected - [CH4] (n)	15
Rejected — [CH4] (%)	1
Model performance	
Passed — Linear model (n)	852
Passed — HM81 model (n)	451
Passed — Polynomial model (n)	858



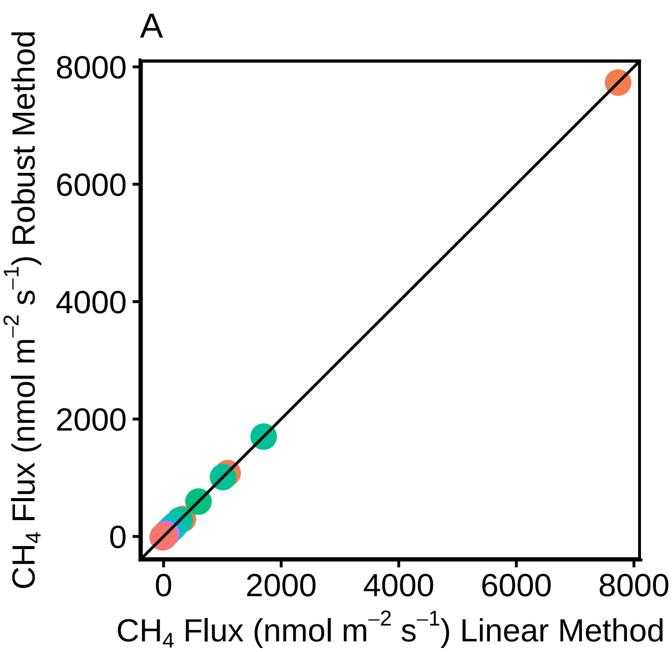
1.4 Remove any fluxes that do not fit criteria

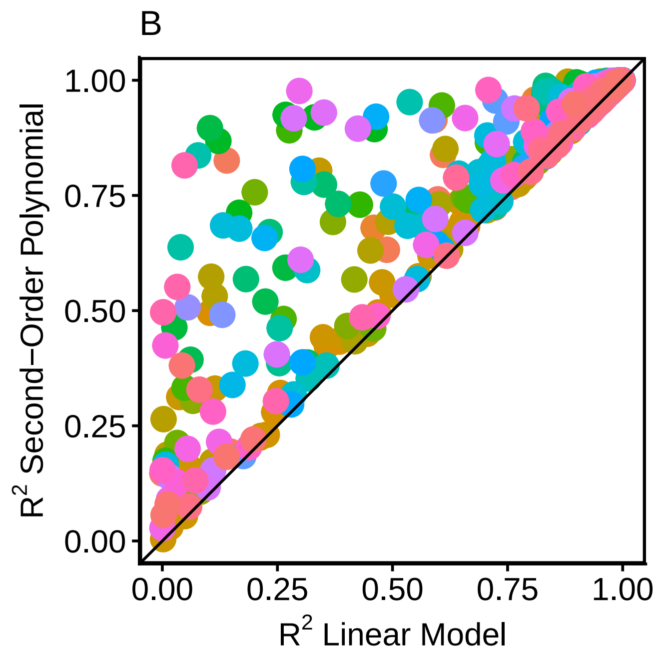
```
                 Plot_ID           TIMESTAMP HM81_AIC HM81_flux.estimate
1 2023_01_05_OWC_TR_58_1 2023-01-05 11:06:46 222.8080        -0.60439179
2 2023_01_05_OWC_TR_58_2 2023-01-05 11:08:12 186.7132        -0.41206877
```

```
3 2023_01_05_OWC_TR_60_1 2023-01-05 11:21:11       NA                NA
4 2023_01_05_OWC_TR_60_2 2023-01-05 11:22:34 156.1166       -0.19504349
5 2023_01_05_OWC_TR_61_1 2023-01-05 11:27:49 172.9428       -0.09611557
6 2023_01_05_OWC_TR_61_2 2023-01-05 11:29:14       NA                NA
  HM81_p.value HM81_r.squared HM81_RMSE  lin_AIC lin_flux.estimate
1 6.269688e-31      0.9397137  2.158799 152.9344     -0.5924959218
2 3.727399e-31      0.9410043  1.504716 162.4619     -0.4129079742
3           NA             NA        NA 128.6229      0.0006416185
4 1.128174e-13      0.6863264  1.108091 143.2382     -0.1169192217
5 1.590632e-04      0.2592945  1.311144 168.4159     -0.0588229511
6           NA             NA        NA 127.7154     -0.0913932579
  lin_flux.std.error lin_int.estimate lin_int.std.error  lin_p.value
1        0.010519076         16450.12         0.3454520 1.665288e-45
2        0.011570574         16431.66         0.3799837 3.243351e-36
3        0.011845482         16448.34         0.3480839 9.570600e-01
4        0.009547017         16454.71         0.3135290 2.225538e-16
5        0.012280419         16468.10         0.4032954 1.642014e-05
6        0.008174343         16475.01         0.2684497 5.791785e-15
  lin_r.squared  lin_RMSE poly_AIC poly_r.squared poly_RMSE  rob_AIC
1  9.850960e-01 1.0733840 136.1049      0.9901739 0.8902985 152.9675
2  9.636775e-01 1.1806806 147.6966      0.9750435 0.9997182 162.7681
3  6.985037e-05 0.9977665 125.6517      0.1465829 0.9445379 128.7124
4  7.575526e-01 0.9741935 145.0097      0.7681212 0.9732146 143.3485
5  3.234091e-01 1.2531143 168.5142      0.3741989 1.2310834 168.4311
6  7.225492e-01 0.8341235 131.5064      0.7237064 0.8502848 127.8054
  rob_converged rob_flux.estimate rob_flux.std.error  rob_RMSE
1          TRUE      -0.590749481        0.010016803 0.8677538
2          TRUE      -0.419000426        0.011353684 0.9777928
3          TRUE       0.002809075        0.012501125 0.9467892
4          TRUE      -0.114050213        0.008415038 0.7184101
5          TRUE      -0.057399172        0.013026729 1.3274822
6          TRUE      -0.092496022        0.008228596 0.8547605
        TIMESTAMP_min       TIMESTAMP_max    soilp_c   soilp_m soilp_t
1 2023-01-05 11:06:22 2023-01-05 11:07:11 0.00500000 0.2020000    6.50
2 2023-01-05 11:07:48 2023-01-05 11:08:37 0.00500000 0.2020000    6.50
3 2023-01-05 11:20:50 2023-01-05 11:21:33 0.00422449 0.2355510    6.20
4 2023-01-05 11:22:10 2023-01-05 11:22:59 0.00500000 0.2340000    6.20
5 2023-01-05 11:27:25 2023-01-05 11:28:14 0.00400000 0.2463455    6.20
6 2023-01-05 11:28:50 2023-01-05 11:29:39 0.00400000 0.2470000    6.12
  CH4_initial CH4_lin_pass CH4_HM81_pass CH4_poly_pass CH4_init_conc
1     2018.27         TRUE          TRUE          TRUE         FALSE
2     2017.63         TRUE          TRUE          TRUE         FALSE
3     2021.23         TRUE         FALSE          TRUE         FALSE
```

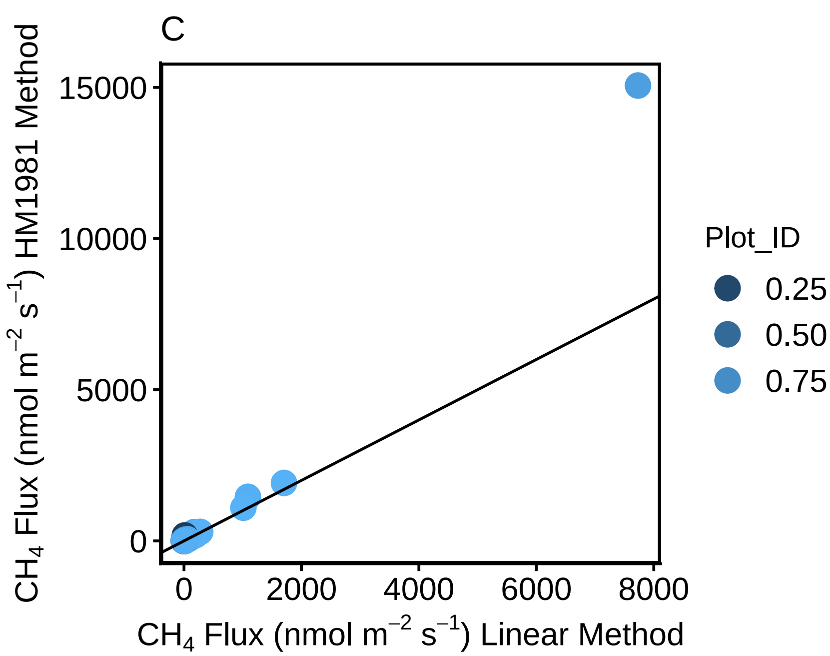
```
4     2020.74          TRUE           TRUE           TRUE          FALSE
5     2024.15          TRUE           TRUE           TRUE          FALSE
6     2023.64          TRUE          FALSE           TRUE          FALSE
  CH4_rejected_model CH4_rejected_manual CH4_flux_rejected
1              FALSE               FALSE             FALSE
2              FALSE               FALSE             FALSE
3              FALSE               FALSE             FALSE
4              FALSE               FALSE             FALSE
5              FALSE               FALSE             FALSE
6              FALSE               FALSE             FALSE
```

1.5 Do some visualization of QAQC plots to assess fluxes






```
Warning: Removed 410 rows containing missing values or values outside the scale range
(`geom_point()`).
```



1.6 Read out final dataframe and export for use elsewhere

```
[1] "TR" "UP" "W"
```

```
 [1] 58 60 61 63 64 50 51 52 55 56 65 66 68 71 72 11 12 14 15  9  2  3  4  6  7
[26] 17 18 20 22 23 34 36 37 38 39 25 26 29 41 42 43 46 47 48 62 67 27 30  8 33
[51] 53 45 70 35 40 59 32 69 28  1 21 44
```

```
[1] 1 2
```